

Multinational Firms and Trade in Latin America

Stylized Facts: All Tables and Figures, All Goods

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Data

The data are the customs export records of nine Latin American countries at the firm, destination, HS6 product and year level, matched to the ownership records of Orbis and Dun & Bradstreet. The years covered are Argentina 2011–2019, Chile 2009–2022, Colombia 2010–2021, Costa Rica 2010–2019, the Dominican Republic 2012–2019, El Salvador 2006–2018, Paraguay 2012–2020, Peru 2010–2019 and Uruguay 2010–2019.

An exporter found in Orbis or Dun & Bradstreet is a multinational affiliate (MNE). It is a *domestic* MNE when its ultimate parent is located in the exporting country and a *foreign* MNE otherwise; matched exporters without a recorded parent country (7.5 percent of foreign-MNE export value) are counted as foreign. Exporters not found in either database are *local* firms. Sectors are defined on HS chapters: agriculture 01–24, mining and fuels 25–27 and 71, manufacturing 28–97 excluding 71.

1 Fact 1: Multinationals Account for a Large Share of Exports

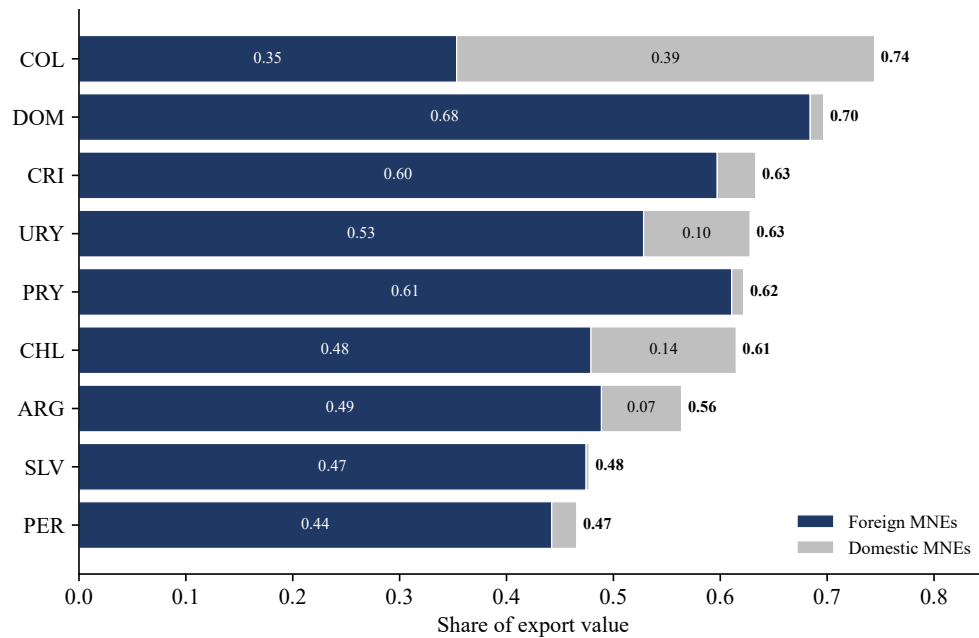


Figure 1: Multinational Share of Export Value, by Exporting Country

Notes: Shares of export value, all available years pooled. The figure at the end of each bar is the total MNE share.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

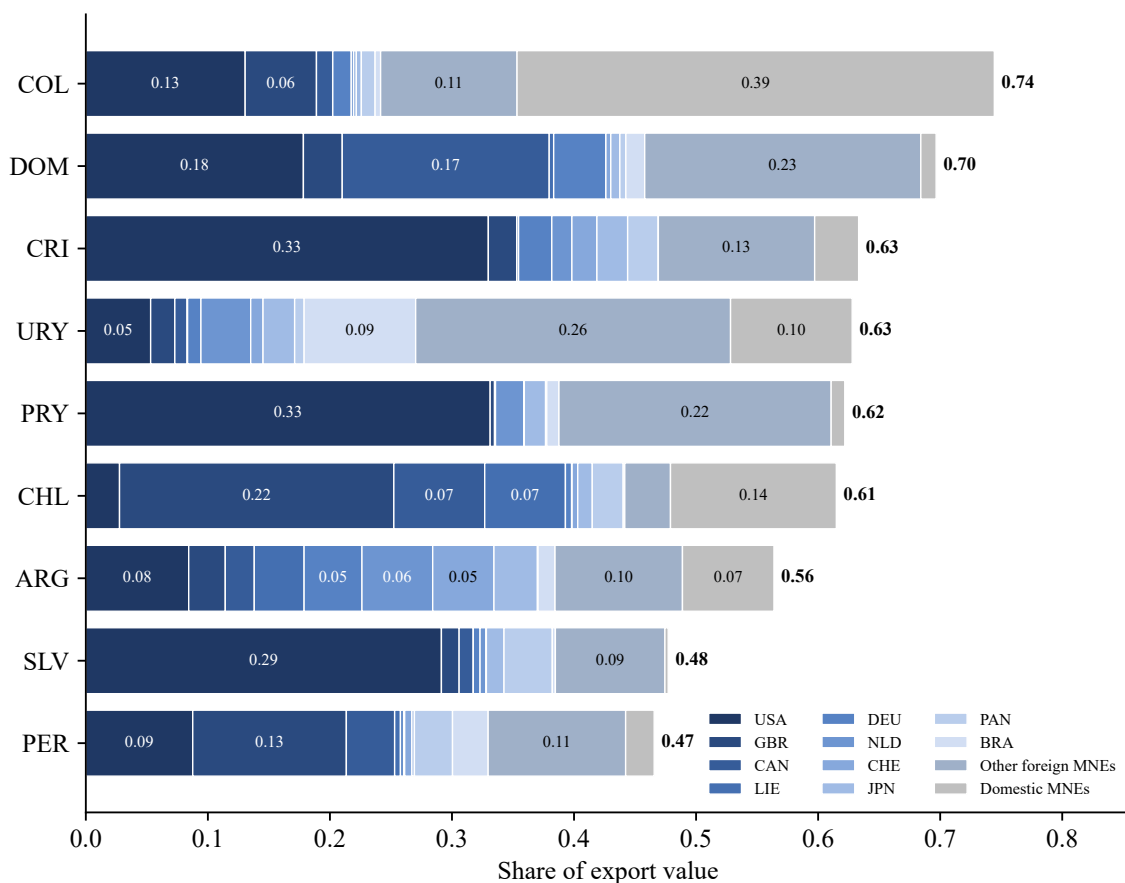
Table 1: Multinational Share of Export Value, by Exporting Country

Origin	Foreign	Domestic	All MNEs	Value (US\$bn/yr)
COL	0.353	0.391	0.744	44.0
DOM	0.684	0.013	0.697	8.6
CRI	0.597	0.036	0.633	10.5
URY	0.528	0.100	0.628	7.0
PRY	0.611	0.011	0.622	7.3
CHL	0.479	0.136	0.615	34.2
ARG	0.489	0.075	0.564	66.4
SLV	0.474	0.003	0.477	5.0
PER	0.442	0.023	0.466	41.4
All	0.463	0.141	0.605	224.5

Notes: Shares of export value, all available years pooled. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

1.1 By Parent Country

**Figure 2: Multinational Share of Export Value, by Exporting Country and Parent Country**

Notes: Shares of export value, all available years pooled. Foreign MNEs split by country of the ultimate parent; the ten largest parent countries are shown separately.

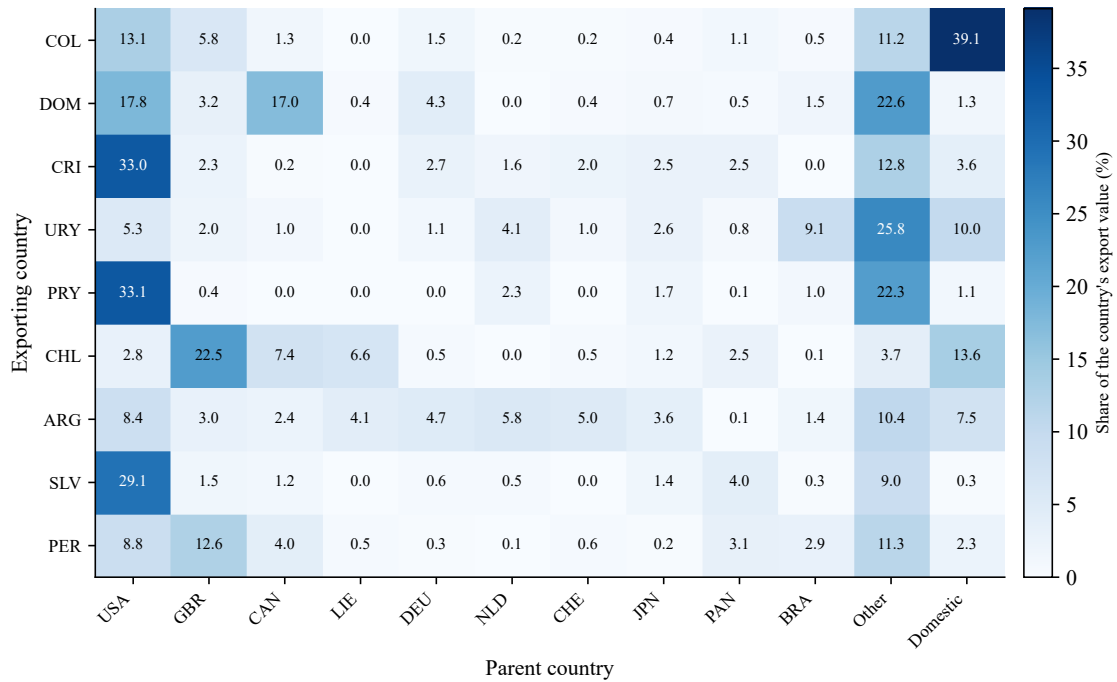
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 2: Multinational Share of Export Value, by Exporting Country and Parent Country

Origin	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Domestic	All MNEs	Value (US\$bn/yr)
COL	0.131	0.058	0.013	0.000	0.015	0.002	0.002	0.004	0.011	0.005	0.112	0.391	0.744	44.0
DOM	0.178	0.032	0.170	0.004	0.043	0.000	0.004	0.007	0.005	0.015	0.226	0.013	0.697	8.6
CRI	0.330	0.023	0.002	0.000	0.027	0.016	0.020	0.025	0.025	0.000	0.128	0.036	0.633	10.5
URY	0.053	0.020	0.010	0.000	0.011	0.041	0.010	0.026	0.008	0.091	0.258	0.100	0.628	7.0
PRY	0.331	0.004	0.000	0.000	0.000	0.023	0.000	0.017	0.001	0.010	0.223	0.011	0.622	7.3
CHL	0.028	0.225	0.074	0.066	0.005	0.000	0.005	0.012	0.025	0.001	0.037	0.136	0.615	34.2
ARG	0.084	0.030	0.024	0.041	0.047	0.058	0.050	0.036	0.001	0.014	0.104	0.075	0.564	66.4
SLV	0.291	0.015	0.012	0.000	0.006	0.005	0.000	0.014	0.040	0.003	0.090	0.003	0.477	5.0
PER	0.088	0.126	0.040	0.005	0.003	0.001	0.006	0.002	0.031	0.029	0.113	0.023	0.466	41.4

Notes: Shares of export value, all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 3: Export Value by Exporting Country and Parent Country**

Notes: Percent of each country's export value, by country of the ultimate parent of the exporter.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

1.2 The United States

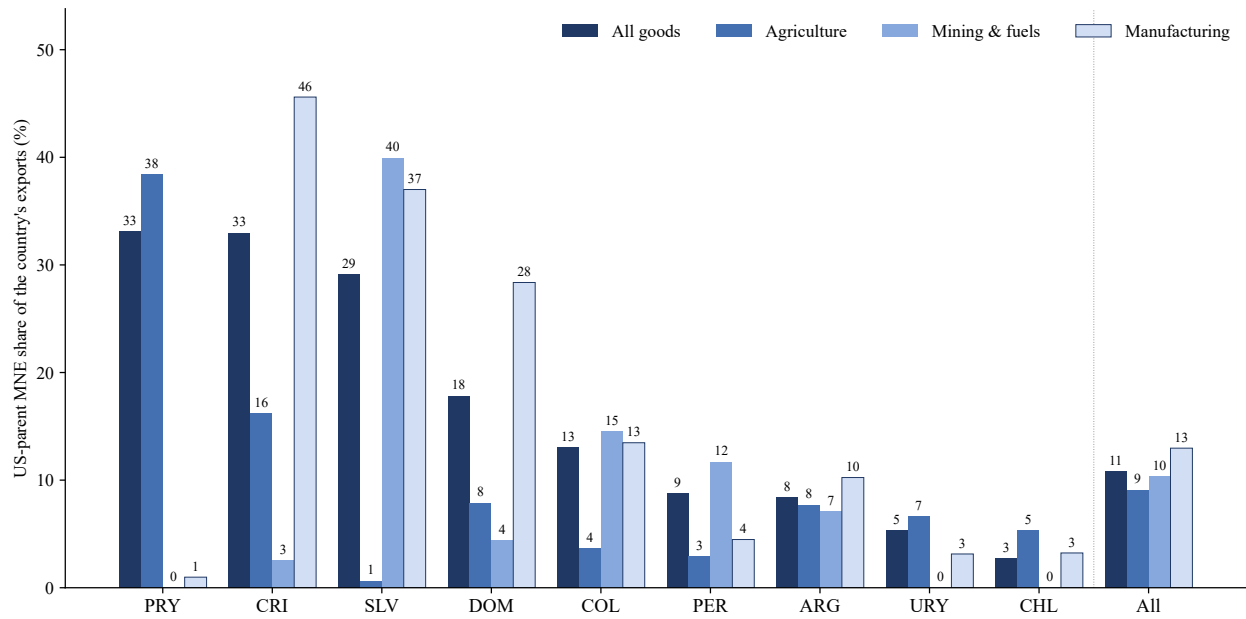


Figure 4: Exports by US-Parent Multinationals, by Exporting Country and Sector

Notes: Percent of each country's exports in each sector shipped by US-parent MNEs, to all destinations.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 3: Exports by US-Parent Multinationals, by Exporting Country and Sector

Origin	US-parent MNE share of the origin's exports in the sector (%)				US-parent MNE exports (US\$bn/yr)			
	All goods	Agriculture	Mining & fuels	Manufacturing	All goods	Agriculture	Mining & fuels	Manufacturing
PRY	33.1	38.5	0.0	1.0	2.2	2.2	0.0	0.0
CRI	33.0	16.2	2.6	45.6	3.2	0.6	0.0	2.5
SLV	29.1	0.6	39.9	37.0	1.4	0.0	0.1	1.3
DOM	17.8	7.9	4.4	28.4	1.5	0.2	0.1	1.2
COL	13.1	3.7	14.6	13.5	5.1	0.2	3.9	1.0
PER	8.8	2.9	11.7	4.5	3.2	0.2	2.8	0.3
ARG	8.4	7.7	7.1	10.2	5.4	2.7	0.5	2.2
URY	5.3	6.6	0.0	3.1	0.3	0.3	0.0	0.1
CHL	2.8	5.4	0.1	3.2	0.9	0.3	0.0	0.5
All	10.8	9.1	10.3	13.0	23.1	6.7	7.3	9.1

Notes: Left panel: percent of each country's exports in the sector shipped by US-parent MNEs. Right panel: exports by US-parent MNEs. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

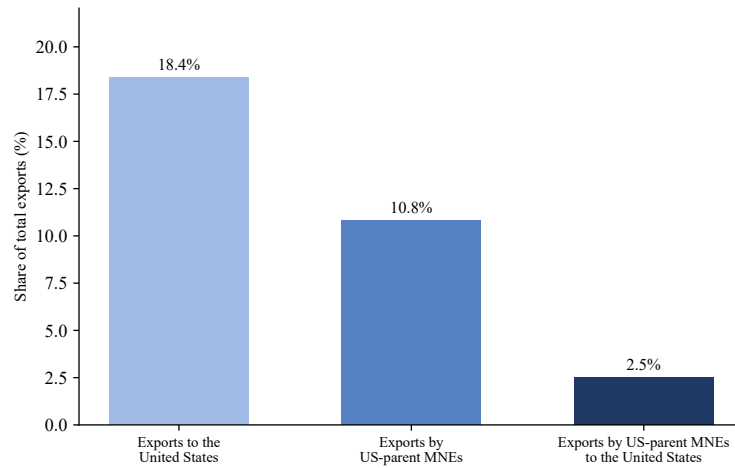


Figure 5: Exports to the United States and Exports by US-Parent Multinationals

Notes: Shares of the total exports of the nine countries. The third bar is exports by US-parent MNEs shipped to the United States.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

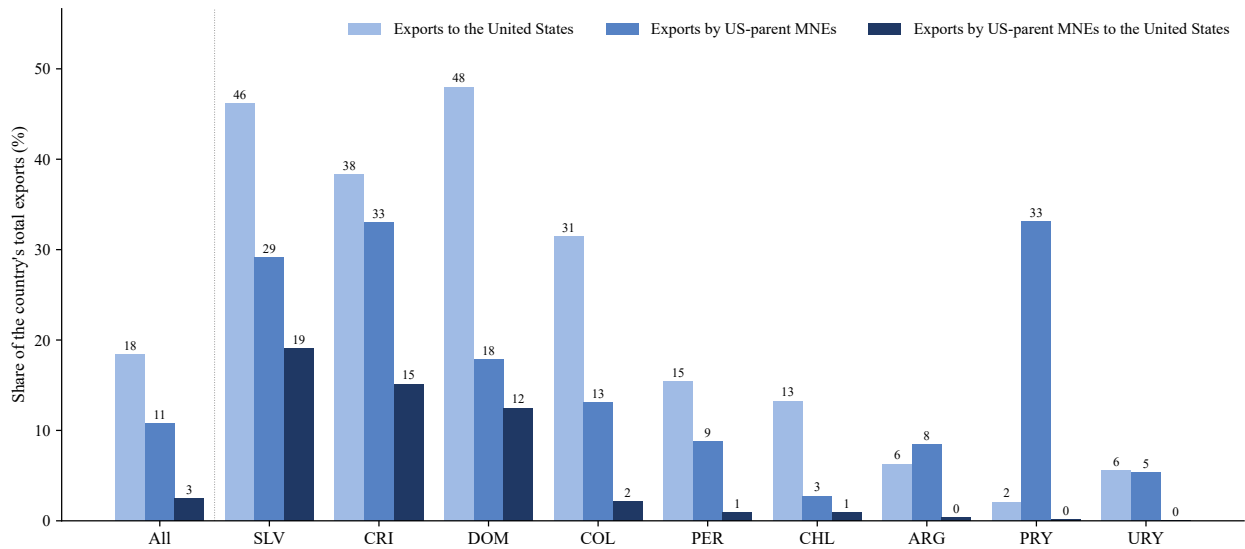


Figure 6: Exports to the United States and Exports by US-Parent Multinationals, by Exporting Country

Notes: Shares of each country's total exports.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 4: Exports to the United States and Exports by US-Parent Multinationals, by Exporting Country

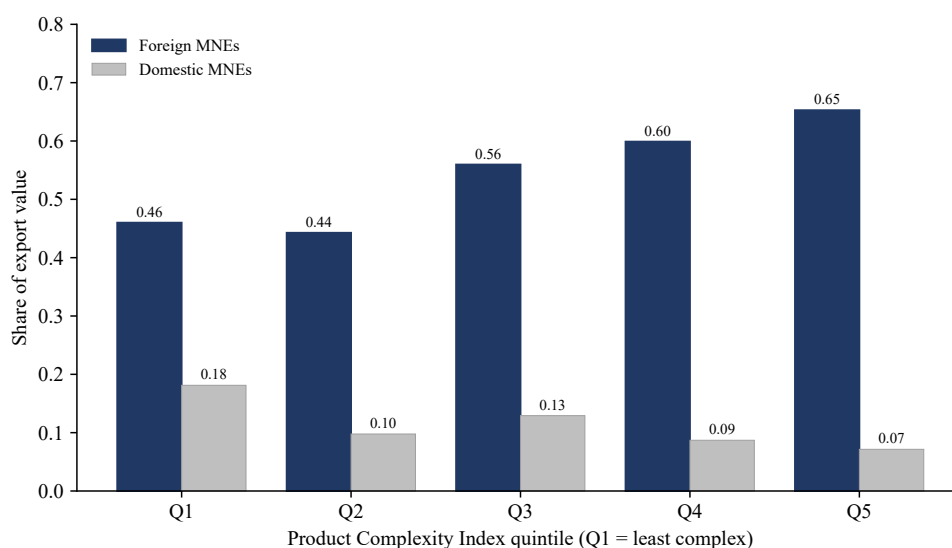
Origin	(1) % of exports going to the USA	(2) % of exports carried by US-parent MNEs	(3) % of exports carried by US-parent MNEs and going to the USA	(4) = (3)/(1): US-parent share of the exports going to the USA (%)	Total exports (US\$bn/yr)
All	18.4	10.8	2.5	13.6	224.5
SLV	46.1	29.1	19.0	41.3	5.0
CRI	38.3	33.0	15.1	39.5	10.5
DOM	48.0	17.8	12.5	26.0	8.6
COL	31.5	13.1	2.1	6.8	44.0
PER	15.4	8.8	0.9	6.0	41.4
CHL	13.3	2.8	0.9	6.9	34.2
ARG	6.3	8.4	0.3	5.5	66.4
PRY	2.1	33.1	0.2	8.1	7.3
URY	5.5	5.3	0.0	0.4	7.0

Notes: Columns (1)–(3): percent of each country’s total exports. Column (4): percent of exports to the United States shipped by US-parent MNEs. Countries sorted by column (3). Values are annual averages in US\$ billion.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

2 Fact 2: Foreign Multinationals Specialize in Complex Products, Domestic Multinationals in Primary Goods

2.1 Product Complexity and Technology

**Figure 7:** Multinational Share of Export Value, by Product Complexity Quintile

Notes: Shares of export value, all available years pooled. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

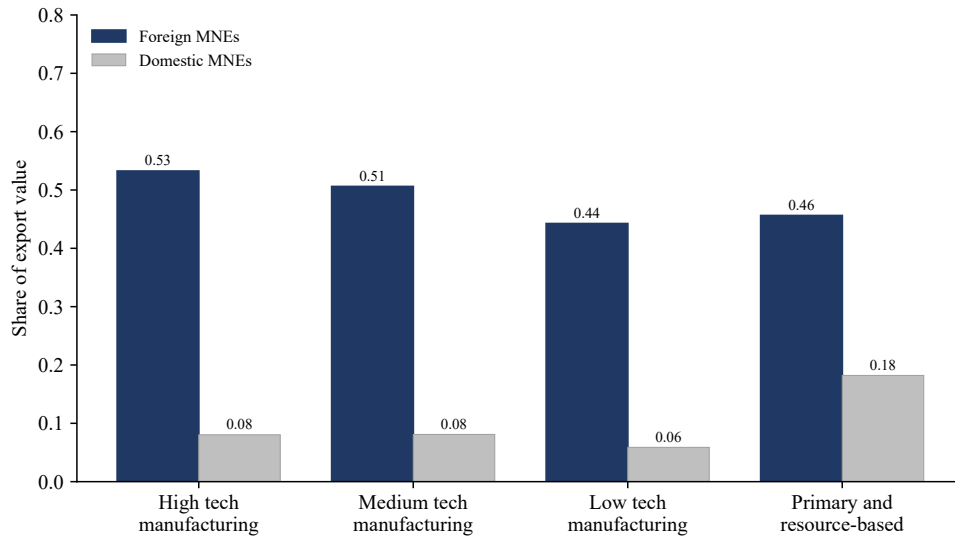


Figure 8: Multinational Share of Export Value, by Technology Class

Notes: Shares of export value, all available years pooled. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

2.2 By Parent Country

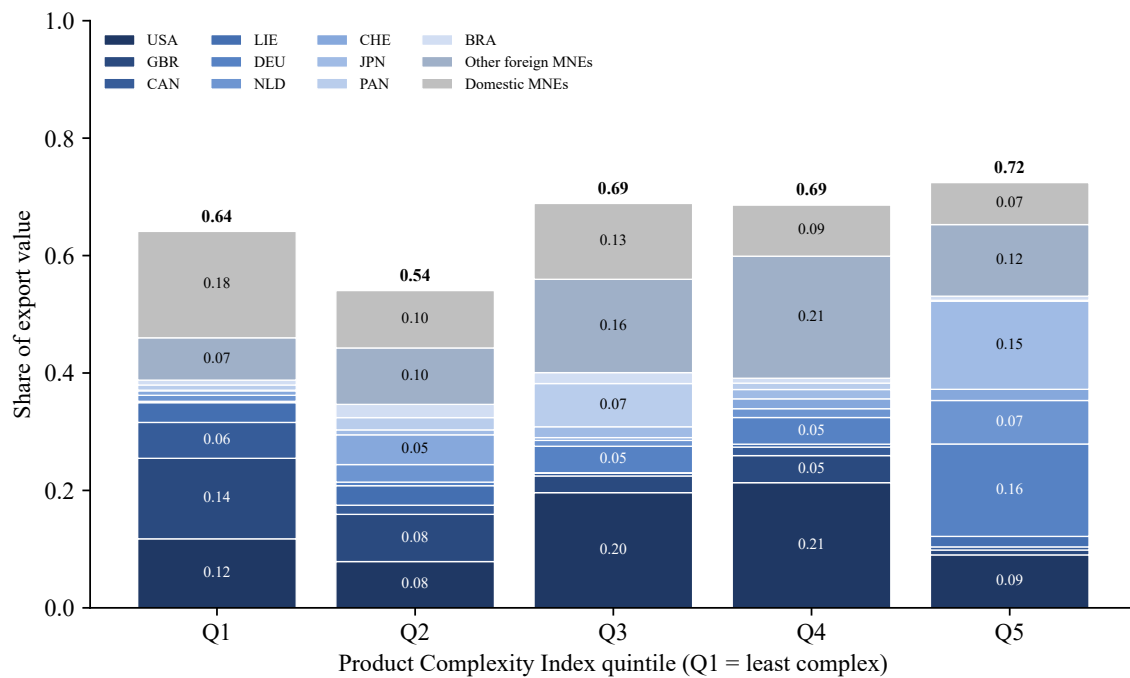


Figure 9: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country

Notes: Shares of export value, all available years pooled. Foreign MNEs split by country of the ultimate parent. Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex).

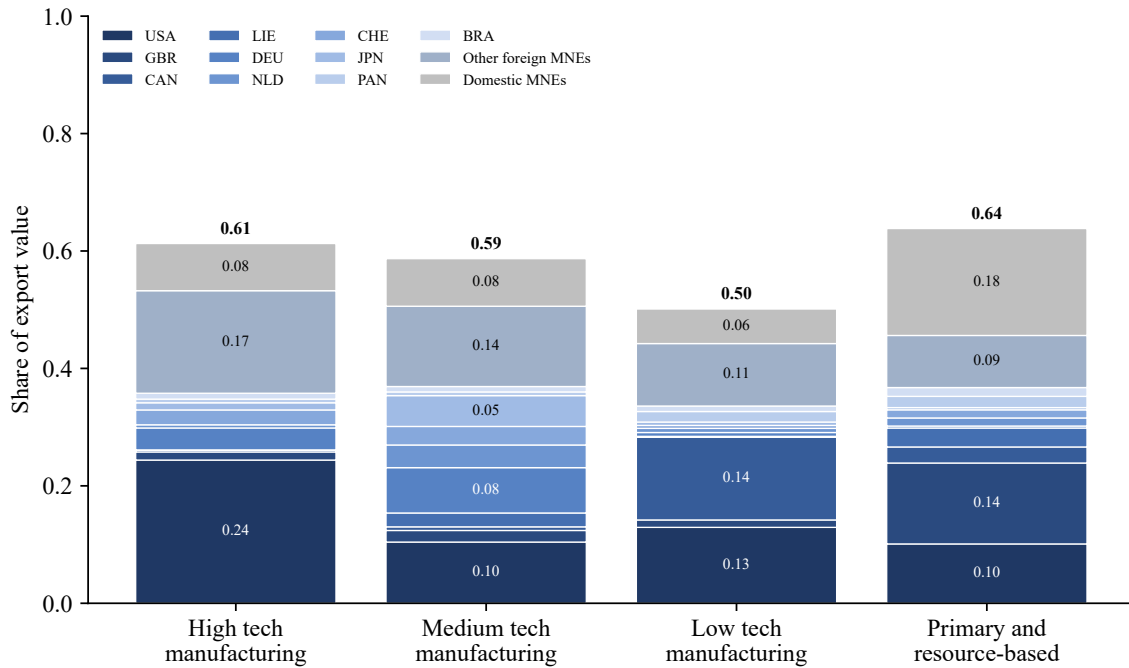
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 5: Multinational Share of Export Value, by Product Complexity Quintile and Parent Country

PCI quintile	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Domestic	All MNEs	Value (US\$bn/yr)
Q1	0.118	0.137	0.061	0.034	0.002	0.010	0.008	0.001	0.009	0.008	0.072	0.181	0.641	106.7
Q2	0.079	0.080	0.016	0.033	0.006	0.030	0.051	0.008	0.021	0.023	0.096	0.098	0.540	44.3
Q3	0.196	0.028	0.005	0.000	0.045	0.010	0.005	0.018	0.074	0.019	0.159	0.129	0.689	13.4
Q4	0.213	0.046	0.014	0.005	0.045	0.015	0.017	0.016	0.011	0.008	0.208	0.087	0.686	13.1
Q5	0.090	0.008	0.005	0.018	0.157	0.074	0.019	0.150	0.002	0.007	0.122	0.071	0.724	14.4

Notes: Shares of export value, all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 10: Multinational Share of Export Value, by Technology Class and Parent Country**

Notes: Shares of export value, all available years pooled. Foreign MNEs split by country of the ultimate parent. Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 6: Multinational Share of Export Value, by Technology Class and Parent Country

Technology category	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other	Domestic	All MNEs	Value (US\$bn/yr)
High tech manufacturing	0.244	0.014	0.003	0.000	0.037	0.005	0.025	0.012	0.006	0.010	0.175	0.080	0.613	6.0
Medium tech manufacturing	0.104	0.020	0.006	0.024	0.077	0.039	0.032	0.053	0.006	0.009	0.137	0.081	0.587	49.5
Low tech manufacturing	0.129	0.013	0.141	0.002	0.007	0.007	0.005	0.006	0.018	0.009	0.106	0.059	0.501	29.8
Primary and resource-based	0.101	0.138	0.027	0.032	0.003	0.014	0.014	0.004	0.019	0.015	0.089	0.182	0.638	130.9

Notes: Shares of export value, all available years pooled. Other: all remaining foreign parents. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

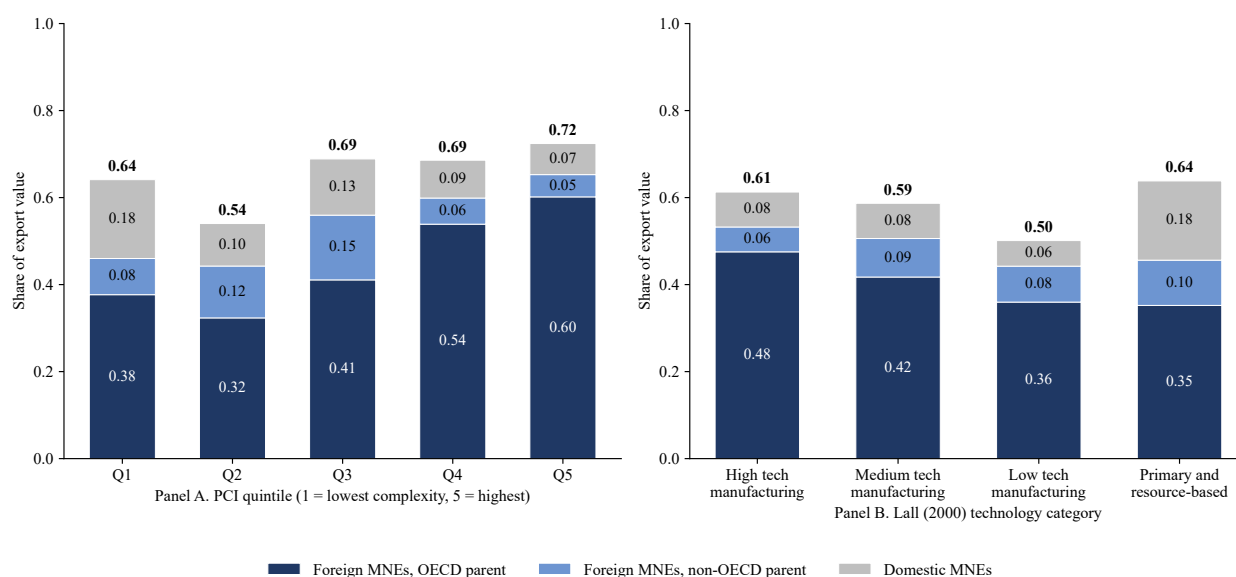


Figure 11: Multinational Share of Export Value, by Product Sophistication and Location of the Parent

Notes: Shares of export value, all available years pooled. Foreign MNEs are split by whether the ultimate parent is in an OECD member country. Panel A: Quintiles of the Hausmann–Hidalgo Product Complexity Index across HS6 products (Q1 = least complex). Panel B: Lall (2000) technology classes: primary and resource-based products, and low-, medium- and high-technology manufactures.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

2.3 Other Measures of Product Sophistication

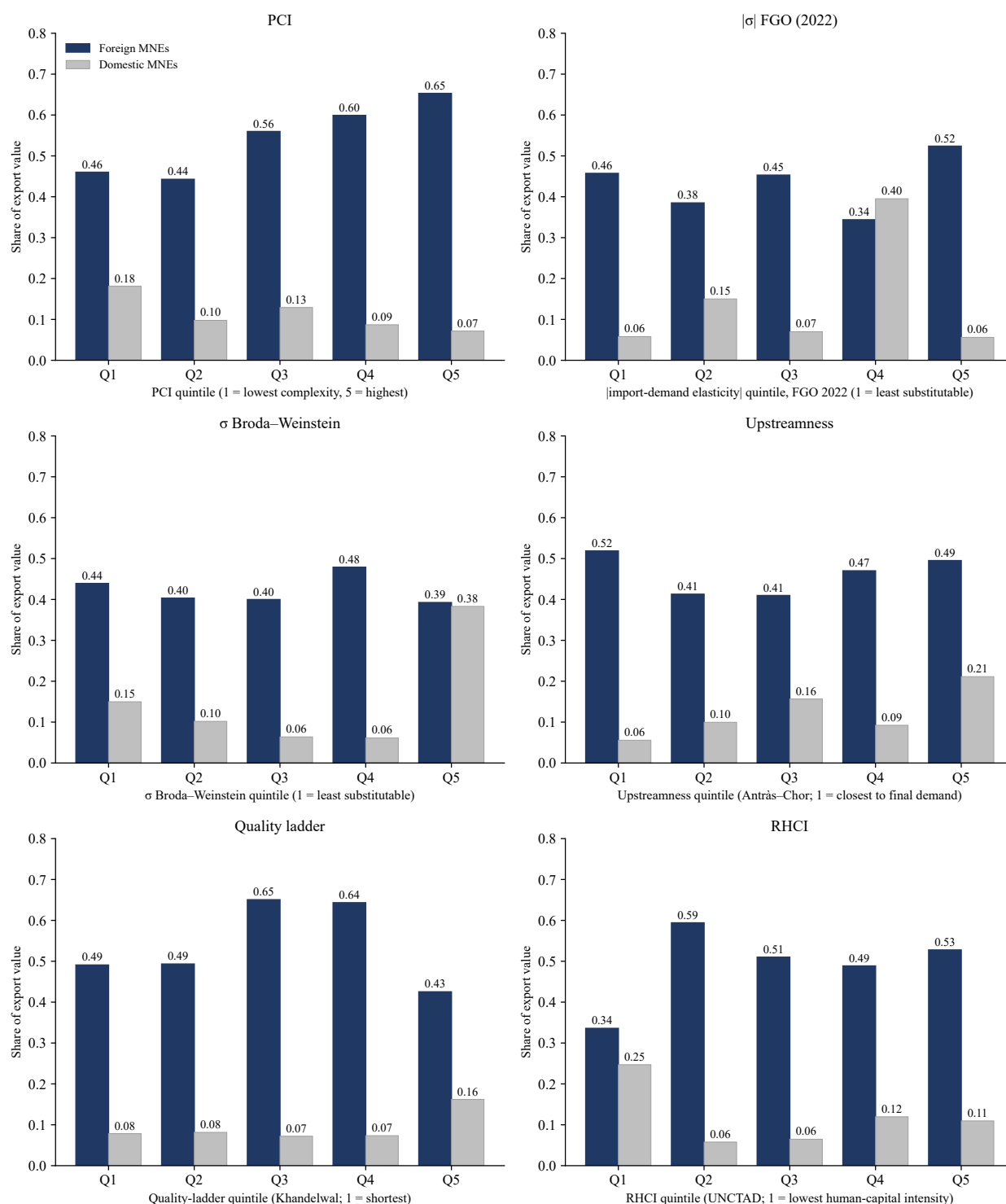


Figure 12: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

Notes: Shares of export value, all available years pooled. Quintiles computed over HS6 products. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

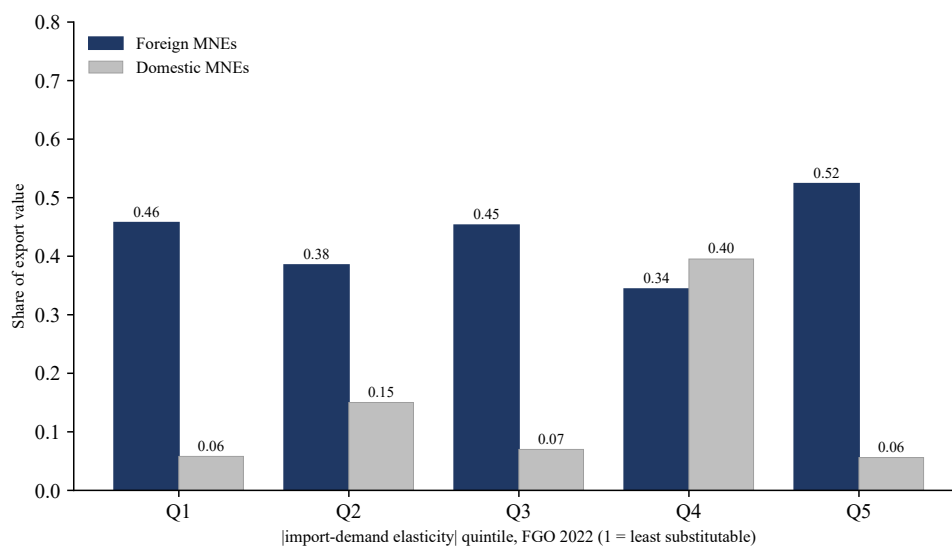


Figure 13: Multinational Share of Export Value, by Quintile of the Import-Demand Elasticity

Notes: Shares of export value, all available years pooled. Quintiles of the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022) across HS6 products; higher quintiles are more substitutable products.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

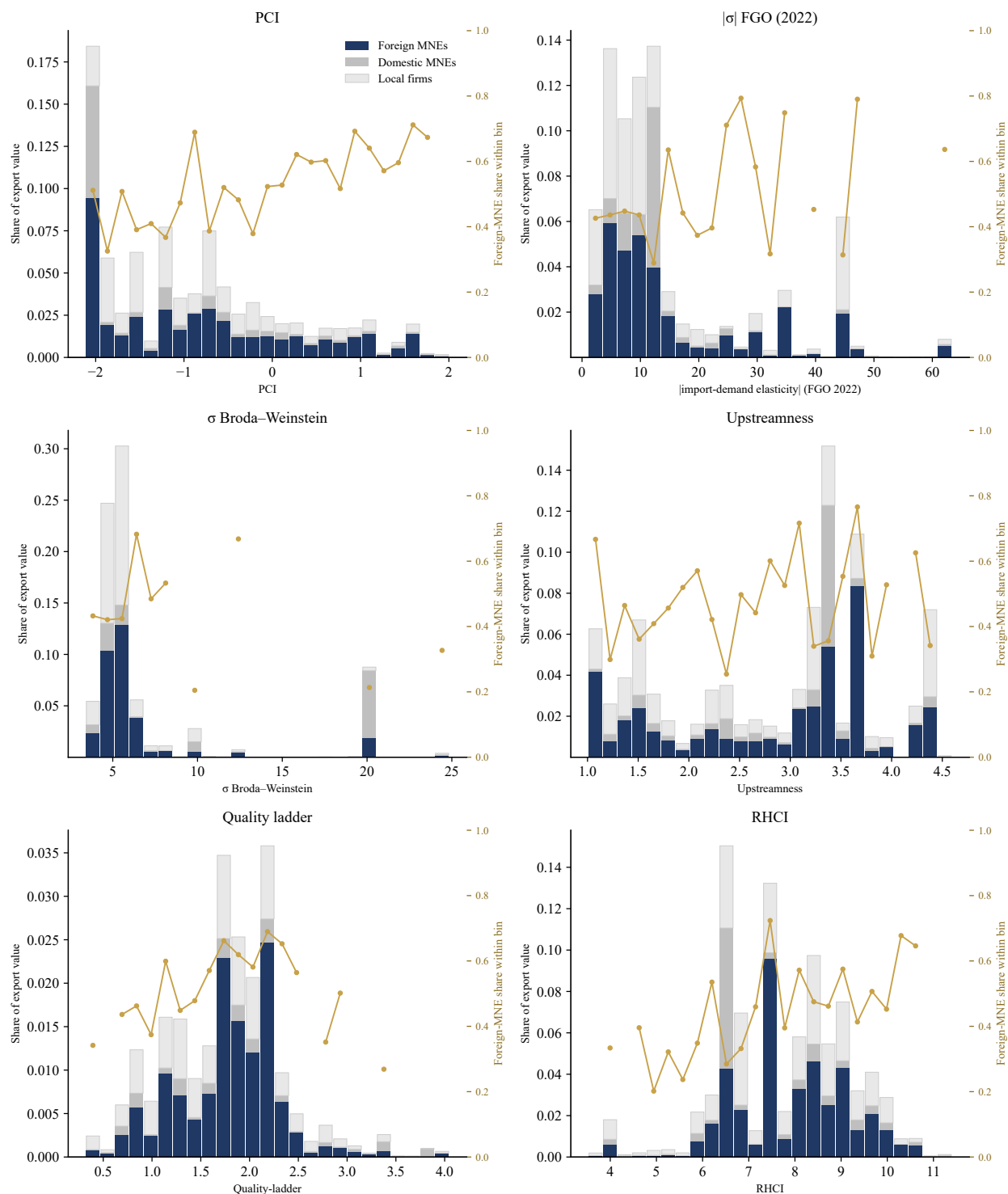


Figure 14: Distribution of Export Value across Six Sophistication Measures

Notes: Bars: export value in each bin, by type of exporter. Line (right axis): foreign-MNE share of each bin. The six measures are the Product Complexity Index, the absolute import-demand elasticity of Fontagné, Guimbard and Orefice (2022), the elasticity of substitution of Broda and Weinstein, upstreamness, the quality ladder and the revealed human-capital intensity of UNCTAD.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

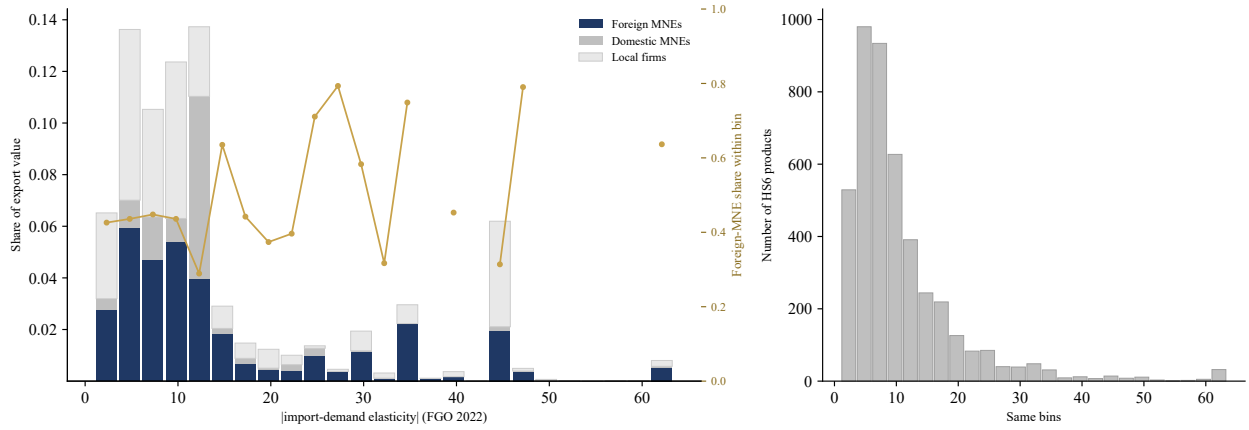


Figure 15: Distribution of Export Value across the Import-Demand Elasticity

Notes: Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

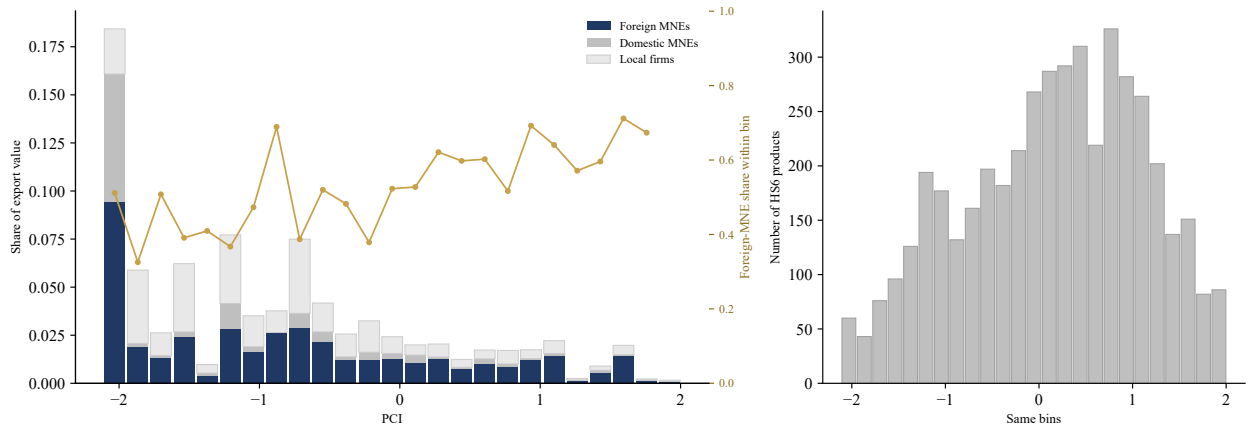


Figure 16: Distribution of Export Value across the Product Complexity Index

Notes: Left: export value in each bin, by type of exporter, and the foreign-MNE share of each bin. Right: number of HS6 products in each bin.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 7: Multinational Share of Export Value, by Quintile of Six Sophistication Measures

Measure	Foreign MNE share					Domestic MNE share				
	Q1	Q2	Q3	Q4	Q5	Q1	Q2	Q3	Q4	Q5
PCI	0.46	0.44	0.56	0.60	0.65	0.18	0.10	0.13	0.09	0.07
$ \sigma $ FGO (2022)	0.46	0.38	0.45	0.34	0.52	0.06	0.15	0.07	0.40	0.06
σ Broda–Weinstein	0.44	0.40	0.40	0.48	0.39	0.15	0.10	0.06	0.06	0.38
Upstreamness	0.52	0.41	0.41	0.47	0.49	0.06	0.10	0.16	0.09	0.21
Quality ladder	0.49	0.49	0.65	0.64	0.43	0.08	0.08	0.07	0.07	0.16
RHCI	0.34	0.59	0.51	0.49	0.53	0.25	0.06	0.06	0.12	0.11

Notes: Shares of export value, all available years pooled. Quintiles computed over HS6 products. –: fewer than 30 HS6 products carry the measure.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 8: Correlations between Sophistication Measures

	PCI	$ \sigma $ FGO (2022)	σ Broda–Weinstein	Upstreamness	Quality ladder	RHCI
PCI	1.00	-0.22	-0.49	-0.49	0.28	0.74
$ \sigma $ FGO (2022)	-0.22	1.00	0.03	0.62	0.12	0.02
σ Broda–Weinstein	-0.49	0.03	1.00	0.20	-0.11	-0.25
Upstreamness	-0.49	0.62	0.20	1.00	0.12	-0.19
Quality ladder	0.28	0.12	-0.11	0.12	1.00	0.34
RHCI	0.74	0.02	-0.25	-0.19	0.34	1.00

Notes: Correlations across HS6 products, weighted by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

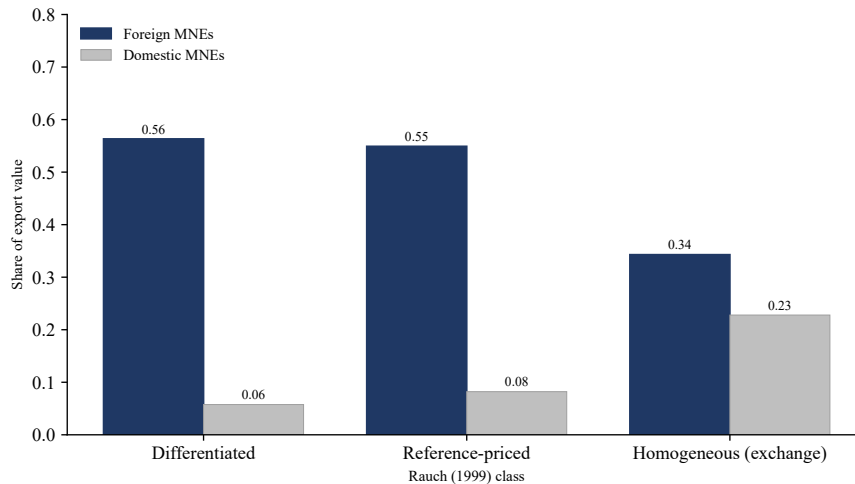
2.4 Rauch Classes and End Use

Table 9: Multinational Share of Export Value, by Rauch Product Class

Rauch (1999) class	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Differentiated	0.563	0.058	0.621	2,712	44.3
Reference-priced	0.549	0.082	0.631	1,636	63.7
Homogeneous (exchange)	0.343	0.228	0.571	319	97.6

Notes: Shares of export value, all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 17:** Multinational Share of Export Value, by Rauch Product Class

Notes: Shares of export value, all available years pooled. Rauch (1999) conservative classification: homogeneous goods are traded on organized exchanges, reference-priced goods have a price quoted in trade publications, and differentiated goods have neither.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 10: Multinational Share of Export Value: Differentiated and Other Products

Rauch (1999): differentiated vs non-differentiated	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Differentiated	0.563	0.058	0.621	2,712	44.3
Non-differentiated	0.426	0.169	0.595	1,955	161.4

Notes: Shares of export value, all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

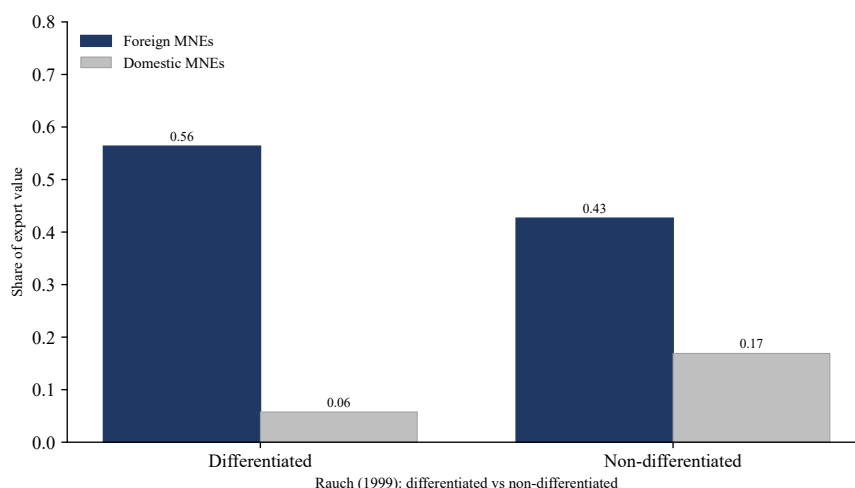


Figure 18: Multinational Share of Export Value: Differentiated and Other Products

Notes: Shares of export value, all available years pooled. Rauch (1999) conservative classification; other products are reference-priced and homogeneous goods.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 11: Multinational Share of Export Value, by End Use

BEC end use	Foreign	Domestic	All MNEs	HS6 products	Value (US\$bn/yr)
Intermediate	0.476	0.155	0.631	3,090	155.0
Consumption	0.438	0.082	0.520	1,201	46.1
Capital	0.647	0.056	0.703	659	9.3

Notes: Shares of export value, all available years pooled. End use from BEC Rev. 4; HS6 products without a BEC correspondence are excluded. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

2.5 Regressions

Table 12: Product Characteristics and Multinational Shares

	(1)	(2)	(3)
<i>Panel A. Dependent variable: share of all MNEs</i>			
Complexity (PCI)	0.0281*** (0.0047)	0.0307*** (0.0045)	0.0409*** (0.0040)
$ \sigma $ FGO (per s.d.)	-0.0078*** (0.0026)	-0.0074*** (0.0025)	-0.0057*** (0.0021)
Upstreamness	0.0087* (0.0047)	0.0079* (0.0044)	0.0108*** (0.0038)
<i>Panel B. Dependent variable: share of foreign MNEs</i>			
Complexity (PCI)	0.0926*** (0.0049)	0.0925*** (0.0047)	0.0855*** (0.0043)
$ \sigma $ FGO (per s.d.)	0.0339*** (0.0023)	0.0331*** (0.0022)	0.0241*** (0.0020)
Upstreamness	-0.0239*** (0.0043)	-0.0253*** (0.0041)	-0.0148*** (0.0034)
<i>Panel C. Dependent variable: share of domestic MNEs</i>			
Complexity (PCI)	-0.0645*** (0.0053)	-0.0618*** (0.0048)	-0.0446*** (0.0038)
$ \sigma $ FGO (per s.d.)	-0.0417*** (0.0027)	-0.0405*** (0.0025)	-0.0298*** (0.0022)
Upstreamness	0.0326*** (0.0041)	0.0332*** (0.0037)	0.0256*** (0.0028)
Origin FE	✓		
Destination FE	✓		
Year FE	✓		
Origin $\times$ year FE		✓	
Destination $\times$ year FE		✓	
Origin $\times$ dest. $\times$ year FE			✓
Observations	1,563,005	1,562,745	1,561,136

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: share of the cell’s export value shipped by each type of MNE. The import-demand elasticity is standardized (mean 0, standard deviation 1). Regressions weighted by the cell’s export value. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 13: Product Sophistication and Multinational Shares

	PCI		Lall		Rauch	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: share of all MNEs</i>						
Complexity (PCI)	-0.0004 (0.0050)	0.0052 (0.0045)				
Lall: high tech (vs primary and resource-based)			-0.0288** (0.0142)	-0.0347** (0.0142)		
Lall: medium tech			-0.0256*** (0.0081)	-0.0111 (0.0074)		
Lall: low tech			-0.0801*** (0.0165)	-0.0797*** (0.0171)		
Rauch: differentiated (vs homogeneous)					0.0819*** (0.0112)	0.0880*** (0.0104)
Rauch: reference-priced					0.1038*** (0.0103)	0.1132*** (0.0104)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
<i>Panel B. Dependent variable: share of foreign MNEs</i>						
Complexity (PCI)	0.0385*** (0.0052)	0.0280*** (0.0050)				
Lall: high tech (vs primary and resource-based)			0.0330** (0.0135)	0.0140 (0.0129)		
Lall: medium tech			0.0400*** (0.0080)	0.0517*** (0.0071)		
Lall: low tech			-0.0000 (0.0118)	-0.0050 (0.0087)		
Rauch: differentiated (vs homogeneous)					0.2088*** (0.0082)	0.1914*** (0.0079)
Rauch: reference-priced					0.1783*** (0.0099)	0.1659*** (0.0099)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
<i>Panel C. Dependent variable: share of domestic MNEs</i>						
Complexity (PCI)	-0.0389*** (0.0058)	-0.0227*** (0.0045)				
Lall: high tech (vs primary and resource-based)			-0.0618*** (0.0143)	-0.0487*** (0.0137)		
Lall: medium tech			-0.0657*** (0.0087)	-0.0628*** (0.0075)		
Lall: low tech			-0.0801*** (0.0148)	-0.0747*** (0.0155)		
Rauch: differentiated (vs homogeneous)					-0.1270*** (0.0101)	-0.1034*** (0.0084)
Rauch: reference-priced					-0.0745*** (0.0086)	-0.0527*** (0.0075)
Observations	1,671,802	1,670,186	1,788,929	1,787,370	1,667,472	1,665,924
Origin × year and destination × year FE	✓		✓		✓	
Origin × dest. × year FE		✓		✓		✓

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: share of the cell’s export value shipped by each type of MNE. Omitted categories: primary and resource-based products (Lall) and homogeneous products (Rauch). Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

3 Fact 3: Multinationals from a Small Set of Countries Dominate Exports

3.1 Parent Countries

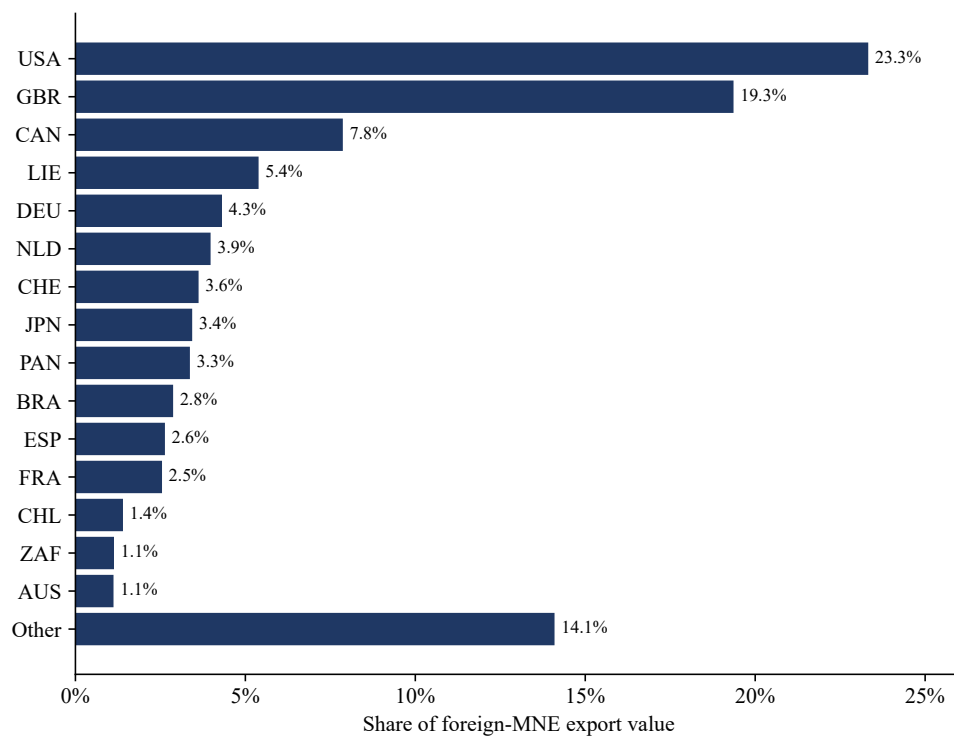


Figure 19: Foreign-Multinational Exports, by Parent Country

Notes: Percent of foreign-MNE export value, by country of the ultimate parent.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 14: Foreign-Multinational Exports, by Parent Country

Parent country	Share (%)	Value (US\$bn/yr)
USA	23.3	23.1
GBR	19.3	16.7
CAN	7.8	7.4
LIE	5.4	4.9
DEU	4.3	4.6
NLD	3.9	4.4
CHE	3.6	3.9
JPN	3.4	3.5
PAN	3.3	3.0
BRA	2.8	3.0
ESP	2.6	2.8
FRA	2.5	2.5
CHL	1.4	1.4
ZAF	1.1	1.0
AUS	1.1	1.1
Other (150 countries)	14.1	14.2
All foreign MNEs	100.0	105.4

Notes: Percent of foreign-MNE export value, by country of the ultimate parent. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

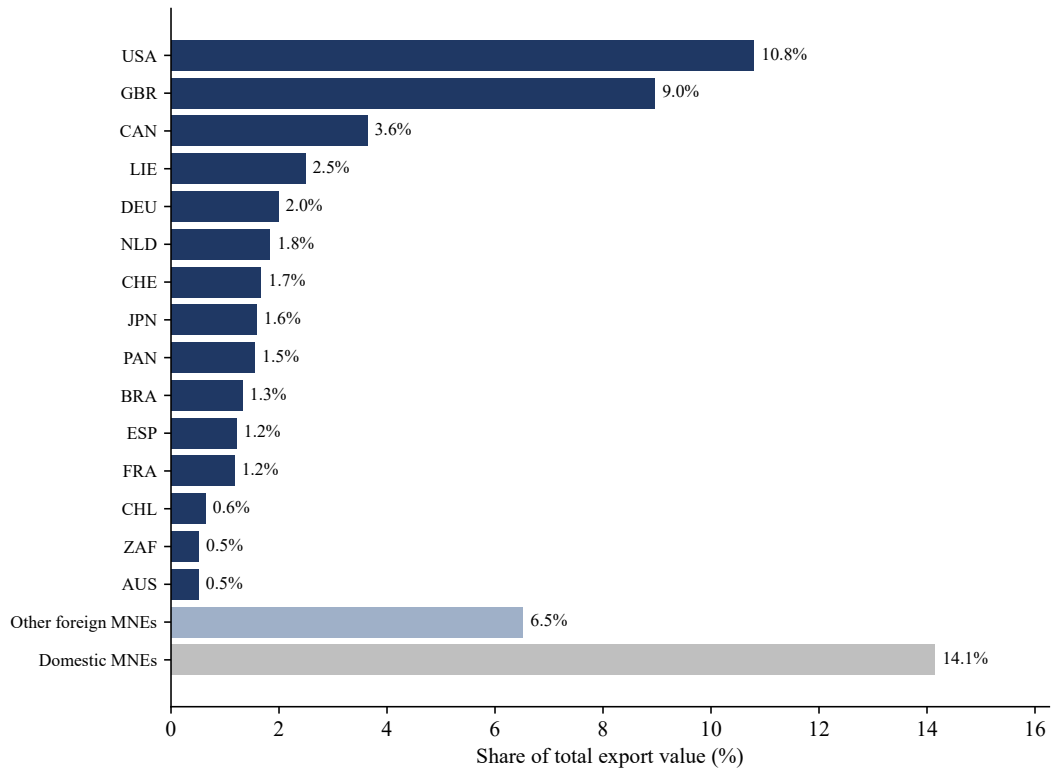


Figure 20: Share of Total Exports, by Parent Country

Notes: Percent of total export value shipped by foreign MNEs of each parent country, other foreign MNEs and domestic MNEs; the remainder is local firms.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 15: Share of Total Exports, by Parent Country

Exporter type / parent country	Share of total exports (%)
USA	10.8
GBR	9.0
CAN	3.6
LIE	2.5
DEU	2.0
NLD	1.8
CHE	1.7
JPN	1.6
PAN	1.5
BRA	1.3
ESP	1.2
FRA	1.2
CHL	0.6
ZAF	0.5
AUS	0.5
Other foreign MNEs	6.5
Domestic MNEs	14.1
All MNEs	60.5
Local firms	39.5

Notes: Percent of total export value, by type of exporter and parent country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

3.2 Parent Region and Destination Region

Table 16: Multinational Exports by Parent Region and Destination Region: Export Value

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	4.8	1.8	1.7	3.1	0.2	0.1	0.0	11.6
North America	9.7	7.3	5.9	7.0	0.6	0.0	0.0	30.5
Europe	13.8	5.5	8.4	16.3	1.5	0.5	0.1	46.0
Asia	3.7	1.3	0.8	1.1	0.1	0.0	0.0	7.0
Africa	0.2	0.1	0.3	0.7	0.0	0.0	0.0	1.3
Oceania	0.2	0.1	0.1	0.7	0.0	0.0	0.0	1.1
Rest of World	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Domestic MNEs	7.6	9.4	3.3	7.8	0.8	0.2	0.1	29.1
Local firms	25.1	18.8	19.7	22.7	2.9	0.6	0.3	90.0
Total	65.0	44.2	40.2	59.3	6.1	1.3	0.5	216.7

Notes: Values are annual averages in US\$ billion. Rows: region of the ultimate parent; columns: destination region.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 17: Multinational Exports by Parent Region and Destination Region: Row Percentages

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Latin America	40.4	15.5	14.7	27.2	1.2	0.7	0.3	100.0
North America	31.3	23.6	20.0	23.1	1.7	0.1	0.1	100.0
Europe	28.0	11.7	17.8	38.4	2.9	1.0	0.1	100.0
Asia	51.0	18.7	11.7	17.0	1.1	0.2	0.3	100.0
Africa	10.9	6.6	26.9	53.2	2.2	0.0	0.2	100.0
Oceania	20.1	5.7	8.4	65.5	0.1	0.2	0.0	100.0
Rest of World	32.6	25.0	20.9	14.3	0.0	7.1	0.0	100.0
Domestic MNEs	25.9	32.9	11.1	27.0	2.3	0.5	0.3	100.0
Local firms	27.1	21.6	21.7	25.7	2.9	0.6	0.4	100.0

Notes: Percent of each group's exports, by destination region (rows add to 100). Domestic MNEs and local firms are shown for comparison.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 18: Multinational Exports by Parent Region and Destination Region: Column Percentages

Parent region / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World
Latin America	7.4	3.9	4.3	5.1	2.6	6.2	6.3
North America	14.9	15.7	15.0	11.2	9.8	1.8	4.9
Europe	20.4	11.9	20.5	28.7	25.0	37.1	10.3
Asia	5.2	2.7	1.9	1.8	1.3	0.9	3.2
Africa	0.2	0.2	0.9	1.1	0.5	0.0	0.4
Oceania	0.3	0.1	0.2	1.1	0.0	0.1	0.0
Rest of World	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Domestic MNEs	13.1	23.1	8.9	13.9	13.5	12.2	15.0
Local firms	38.3	42.4	48.3	37.1	47.3	41.4	59.9
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: Percent of the exports reaching each destination region, by parent region (columns add to 100).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

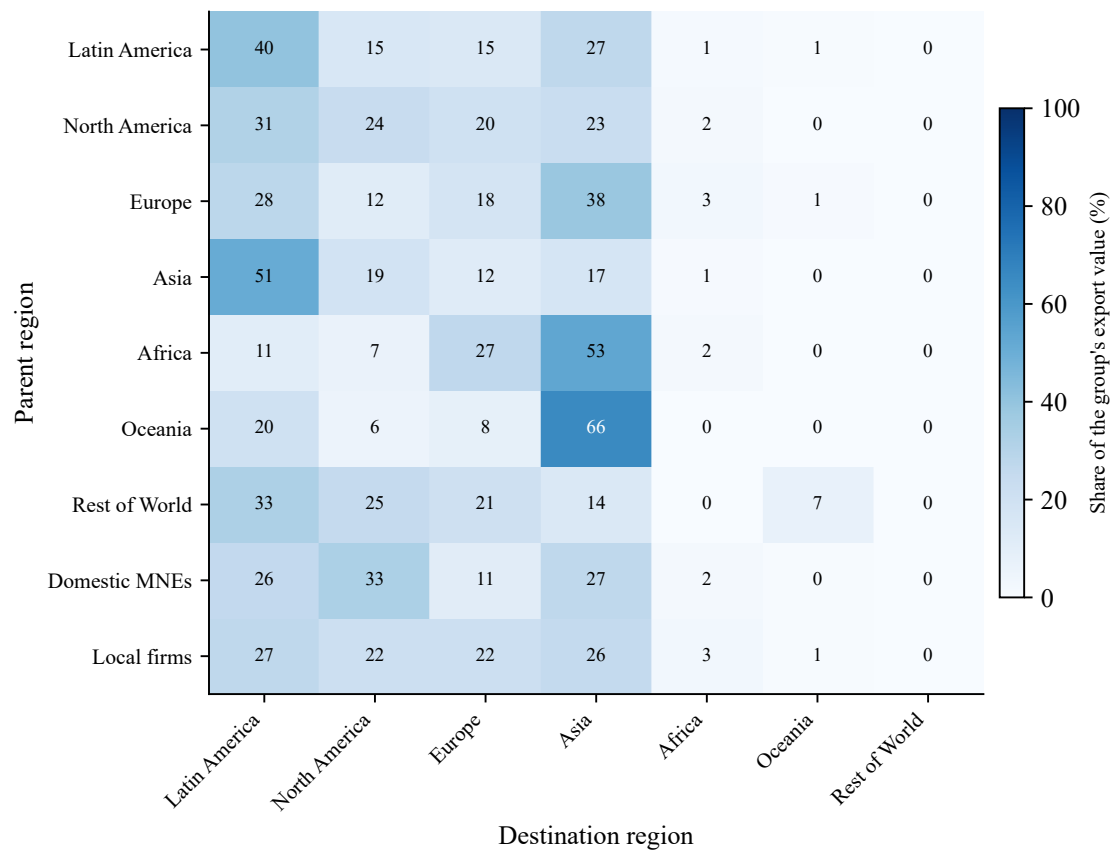


Figure 21: Multinational Exports by Parent Region and Destination Region

Notes: Percent of each group's exports, by destination region (rows add to 100).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 19: Foreign-Multinational Exports by Exporting Country and Destination Region

Origin	Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
ARG		45.5	9.0	15.5	23.0	6.2	0.8	0.0	32.5
CHL		13.1	11.6	13.3	60.5	0.3	1.2	0.0	16.4
COL		40.7	19.7	25.4	12.6	0.5	0.3	0.9	15.5
CRI		33.0	40.7	16.2	9.8	0.1	0.1	0.0	6.3
DOM		20.3	59.3	9.4	10.5	0.2	0.2	0.1	5.9
PER		15.4	12.9	25.3	44.9	1.0	0.4	0.0	18.3
PRY		79.4	1.3	11.6	6.6	1.1	0.0	0.0	4.4
SLV		46.8	48.1	4.3	0.6	0.1	0.1	0.0	2.4
URY		39.1	7.7	20.5	28.0	3.0	0.1	1.6	3.7

Notes: Percent of each country's foreign-MNE exports, by destination region (rows add to 100). Last column: foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 20: Local-Firm Exports by Exporting Country and Destination Region

Origin	Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
ARG		32.2	8.3	21.4	28.5	8.4	1.3	0.0	29.0
CHL		15.5	18.8	20.4	44.3	0.6	0.5	0.0	13.2
COL		32.2	32.8	23.5	8.5	0.8	0.3	1.9	11.2
CRI		32.1	35.4	26.3	5.4	0.3	0.5	0.0	3.8
DOM		31.9	46.8	15.7	4.6	0.6	0.0	0.3	2.6
PER		18.3	28.0	24.6	28.1	0.3	0.4	0.2	22.1
PRY		64.6	3.6	20.0	10.6	1.1	0.0	0.0	2.7
SLV		44.9	46.3	4.6	3.7	0.1	0.3	0.1	2.6
URY		38.7	5.2	21.6	26.0	7.3	0.1	1.3	2.6

Notes: Percent of each country's exports by local firms, by destination region (rows add to 100). Last column: local-firm exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

3.3 Parent Country and Destination Country

Table 21: Foreign-Multinational Exports by Parent Country and Destination: Export Value

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations	Total
USA	2.4	5.0	1.9	0.9	0.9	0.7	0.5	0.1	0.8	1.3	8.7	23.1
GBR	5.9	0.9	0.7	1.1	0.5	0.6	1.1	0.6	0.5	0.0	4.7	16.7
CAN	0.8	0.5	0.3	0.3	0.2	0.0	0.1	1.7	0.1	0.0	3.4	7.4
LIE	1.1	0.2	0.2	0.7	0.2	0.2	0.2	0.0	0.1	0.0	2.0	4.9
DEU	0.0	0.7	1.5	0.0	0.1	0.1	0.0	0.0	0.0	0.0	1.9	4.6
NLD	0.4	0.2	1.2	0.0	0.0	0.2	0.1	0.0	0.1	0.1	2.1	4.4
CHE	0.2	0.2	0.3	0.0	0.1	0.1	0.0	0.0	0.1	0.0	2.8	3.9
JPN	0.1	0.3	1.7	0.2	0.0	0.1	0.0	0.0	0.0	0.1	1.0	3.5
PAN	1.0	0.3	0.0	0.1	0.1	0.0	0.2	0.0	0.1	0.0	1.2	3.0
BRA	0.4	0.4	0.6	0.0	0.0	0.1	0.0	0.0	0.1	0.1	1.3	3.0
Other parents	2.2	4.5	2.8	0.5	0.8	1.0	0.3	0.4	0.5	0.9	9.1	23.0
Total	14.6	13.1	11.2	4.0	3.0	3.1	2.5	2.9	2.3	2.5	38.3	97.6

Notes: Values are annual averages in US\$ billion. Ten largest parent countries and ten largest destinations by foreign-MNE export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 22: Foreign-Multinational Exports by Parent Country and Destination: Row Percentages

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations	Total
USA	10.1	22.3	7.8	3.8	4.0	2.9	2.1	0.5	3.5	4.9	38.2	100.0
GBR	36.5	5.6	3.9	7.1	3.0	3.4	6.5	3.3	3.2	0.2	27.3	100.0
CAN	12.8	6.6	4.2	5.3	3.1	0.5	1.9	19.5	0.9	0.2	45.0	100.0
LIE	24.9	3.7	3.7	18.7	3.5	2.5	5.4	0.0	1.8	0.0	35.6	100.0
DEU	1.0	14.8	31.9	1.1	1.7	3.1	0.6	1.1	1.0	0.9	42.8	100.0
NLD	9.2	3.8	26.9	0.4	1.0	3.6	1.1	0.2	2.2	3.1	48.4	100.0
CHE	6.2	5.0	6.8	0.5	2.9	3.5	0.8	0.4	2.0	0.5	71.4	100.0
JPN	2.4	8.7	45.0	8.9	0.4	1.4	0.4	0.5	0.9	1.9	29.5	100.0
PAN	35.0	9.5	1.2	1.8	1.7	1.1	6.2	0.6	2.0	0.2	40.7	100.0
BRA	12.9	14.1	18.6	1.6	0.6	3.1	0.6	0.6	2.1	2.3	43.4	100.0
Other parents	9.5	19.5	11.7	2.2	3.7	4.0	1.6	1.6	2.1	3.7	40.3	100.0

Notes: Percent of each parent country's exports, by destination (rows add to 100). Ten largest parent countries and destinations.

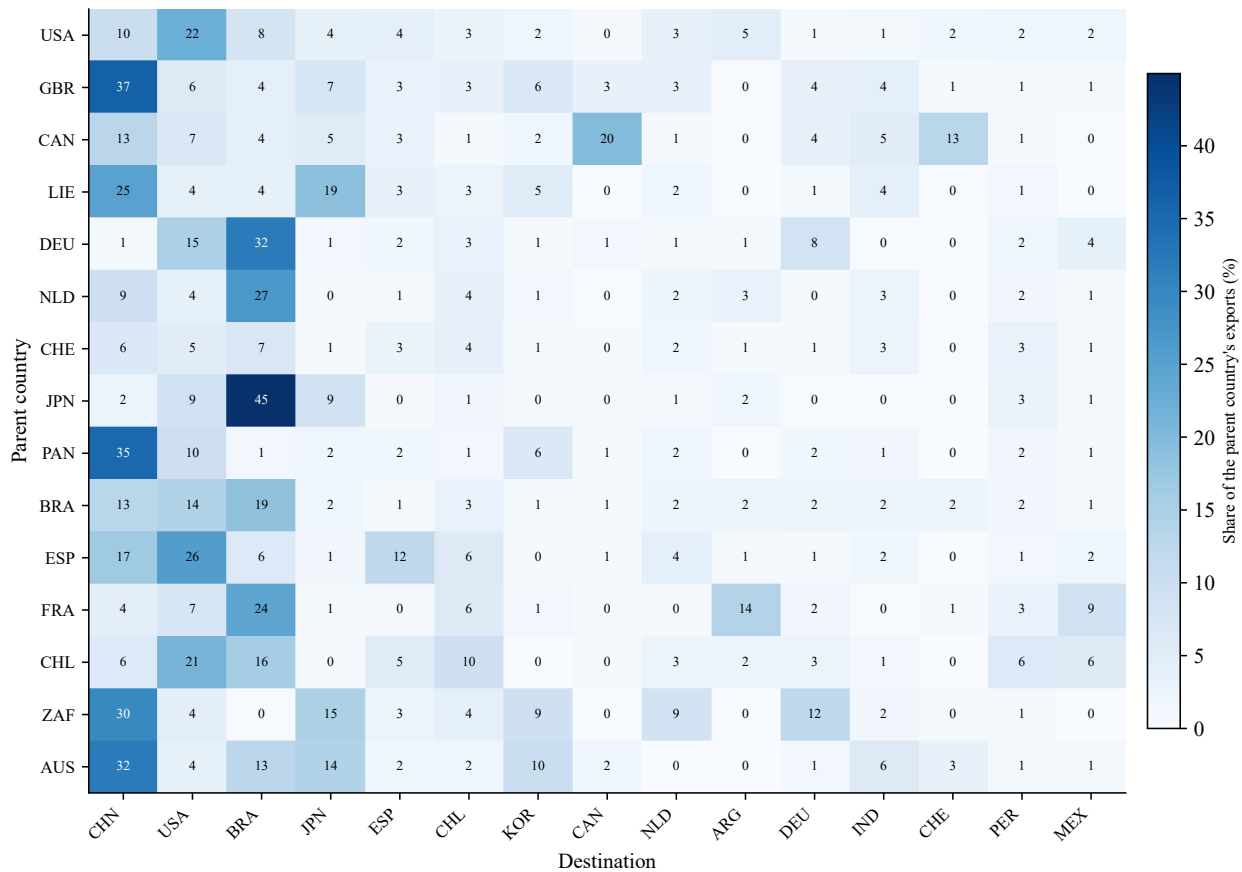
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 23: Foreign-Multinational Exports by Parent Country and Destination: Column Percentages

Parent / Destination	CHN	USA	BRA	JPN	ESP	CHL	KOR	CAN	NLD	ARG	Other destinations
USA	14.6	38.7	17.2	18.8	30.1	22.5	17.0	4.1	33.0	48.1	23.0
GBR	43.7	8.1	7.2	29.3	18.8	21.8	43.5	22.8	25.7	1.5	13.7
CAN	6.2	3.8	3.1	9.0	7.8	1.4	5.1	55.1	3.0	0.8	9.1
LIE	8.3	1.5	1.9	21.5	6.1	4.6	10.1	0.1	4.0	0.1	4.9
DEU	0.3	4.7	13.0	1.0	2.4	4.5	0.9	1.7	1.8	1.6	4.7
NLD	2.3	1.1	10.1	0.3	1.3	4.7	1.6	0.2	3.5	5.2	4.9
CHE	1.4	1.3	2.3	0.4	3.3	4.3	1.0	0.6	2.9	0.8	6.6
JPN	0.5	2.2	14.6	6.5	0.5	1.6	0.4	0.6	1.2	2.7	2.6
PAN	7.2	2.4	0.4	1.3	1.8	1.2	7.2	0.8	2.8	0.2	3.5
BRA	2.3	3.0	5.0	1.0	0.6	2.9	0.6	0.6	2.5	2.8	3.2
Other parents	13.4	33.1	25.3	10.9	27.3	30.4	12.6	13.4	19.6	36.0	23.7
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Notes: Percent of the foreign-MNE exports reaching each destination, by parent country (columns add to 100). Ten largest parent countries and destinations.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 22:** Destination of Foreign-Multinational Exports: Fifteen Largest Parent Countries and Destinations

Notes: Percent of each parent country's exports going to each destination (rows add to 100 including other destinations).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

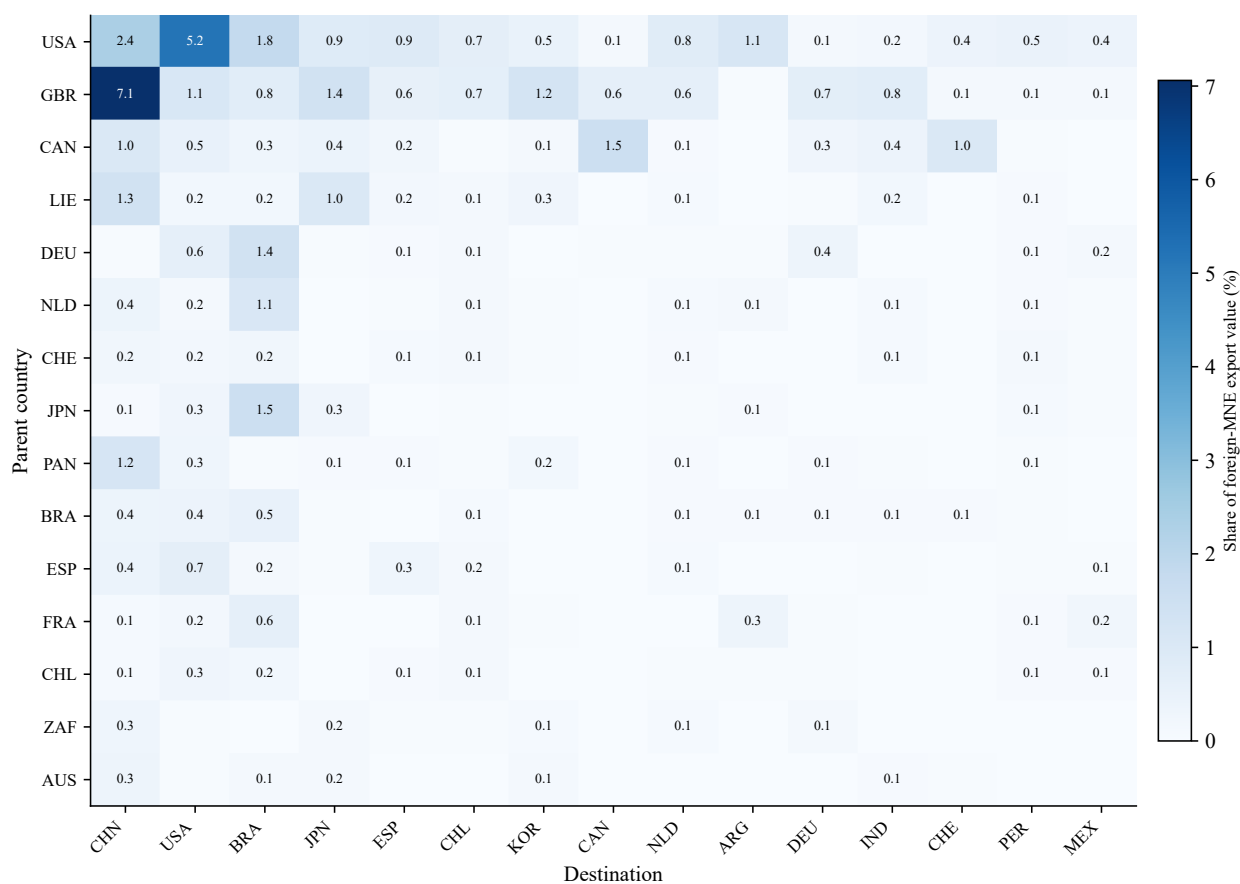


Figure 23: Foreign-Multinational Exports by Parent Country and Destination: Share of the Total

Notes: Percent of all foreign-MNE export value in each parent country and destination pair; fifteen largest parent countries and destinations.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

3.4 Exports to the Parent Country

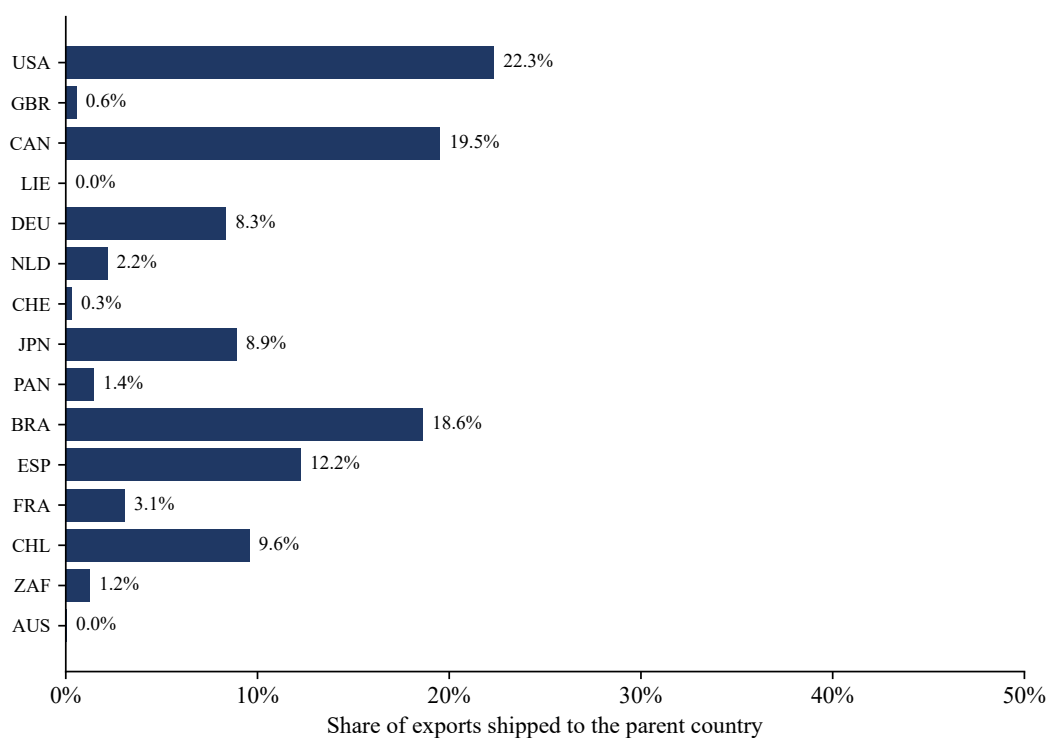


Figure 24: Share of Foreign-Multinational Exports Shipped to the Parent Country

Notes: Percent of each parent country's exports shipped to that same country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 24: Foreign-Multinational Exports to the Parent Countries, by Parent Country

Parent / Destination	USA	GBR	CAN	LIE	DEU	NLD	CHE	JPN	PAN	BRA	Other destinations	Total
USA	22.3	1.0	0.5	0.0	0.6	3.5	1.7	3.8	2.7	7.8	56.2	100.0
GBR	5.6	0.6	3.3	0.0	3.6	3.2	0.6	7.1	0.2	3.9	71.8	100.0
CAN	6.6	2.6	19.5	0.0	3.5	0.9	12.9	5.3	2.6	4.2	41.8	100.0
LIE	3.7	0.4	0.0	0.0	0.9	1.8	0.2	18.7	0.1	3.7	70.5	100.0
DEU	14.8	0.3	1.1	0.0	8.3	1.0	0.0	1.1	0.8	31.9	40.6	100.0
NLD	3.8	0.5	0.2	0.0	0.4	2.2	0.0	0.4	0.4	26.9	65.2	100.0
CHE	5.0	1.0	0.4	0.0	0.7	2.0	0.3	0.5	1.7	6.8	81.6	100.0
JPN	8.7	0.3	0.5	0.0	0.2	0.9	0.0	8.9	1.0	45.0	34.5	100.0
PAN	9.5	0.8	0.6	0.0	1.7	2.0	0.2	1.8	1.4	1.2	80.7	100.0
BRA	14.1	0.5	0.6	0.0	2.3	2.1	2.2	1.6	0.6	18.6	57.4	100.0
Other parents	19.5	0.9	1.6	0.0	2.2	2.1	1.2	2.2	1.7	11.7	56.8	100.0

Notes: Percent of each parent country's exports, by destination (rows add to 100). Destinations are the ten largest parent countries; the diagonal is the share shipped to the parent's own country.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

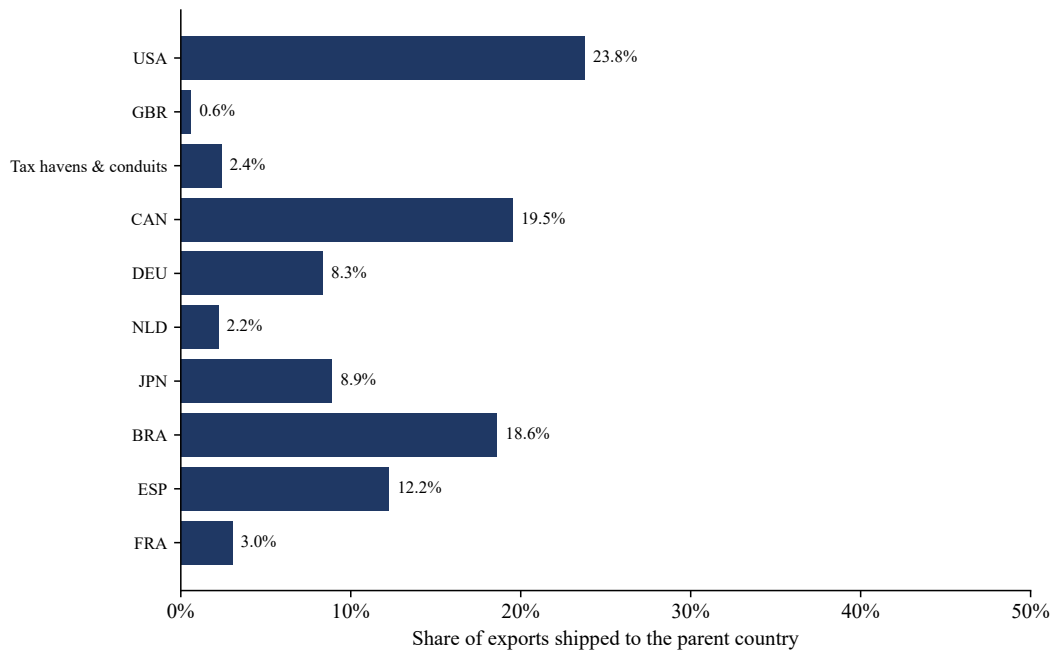


Figure 25: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents
Notes: Percent of each parent's exports shipped to its own jurisdiction. Dependencies are assigned to their sovereign state and tax havens and conduit jurisdictions are pooled.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 25: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction, Consolidated Parents

Consolidated parent	Home share (%)	Exports (US\$bn/yr)
USA	23.8	23.1
GBR	0.6	18.4
Tax havens & conduits	2.4	14.7
CAN	19.5	7.4
DEU	8.3	4.6
NLD	2.2	4.4
JPN	8.9	3.5
BRA	18.6	3.0
ESP	12.2	2.8
FRA	3.0	2.6

Notes: Home share: percent of the parent's exports shipped to its own jurisdiction. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

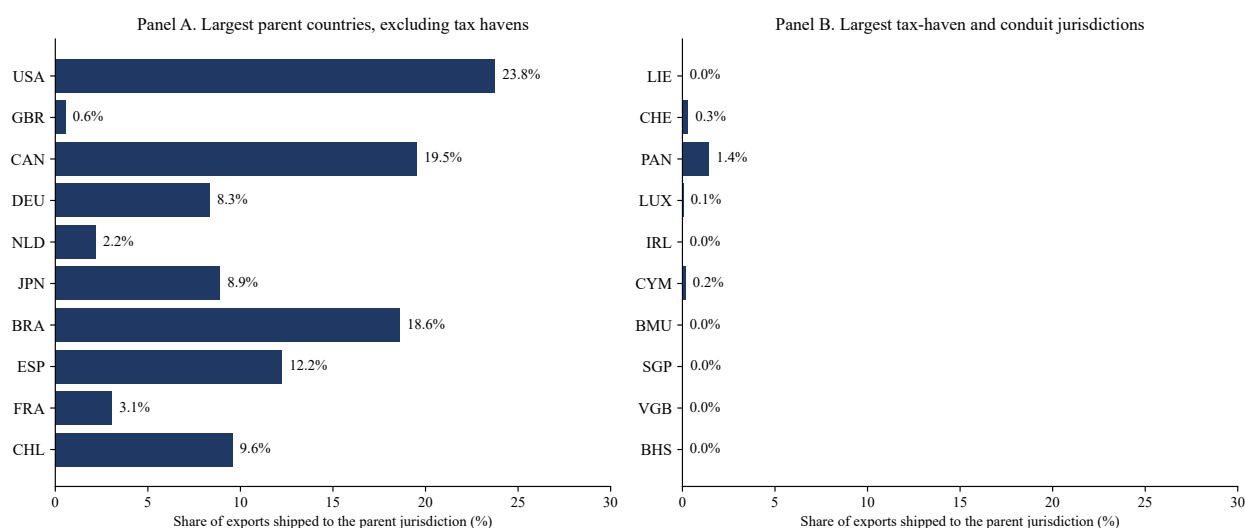


Figure 26: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

Notes: Percent of each parent jurisdiction's exports shipped to that same jurisdiction. Panel A: largest parent countries, excluding tax havens; dependencies are assigned to their sovereign state. Panel B: largest tax-haven and conduit jurisdictions.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 26: Share of Foreign-Multinational Exports Shipped to the Parent Jurisdiction

Panel	Parent	Home share (%)	Exports (US\$bn/yr)
Panel A	USA	23.8	23.1
Panel A	GBR	0.6	16.7
Panel A	CAN	19.5	7.4
Panel A	DEU	8.3	4.6
Panel A	NLD	2.2	4.4
Panel A	JPN	8.9	3.5
Panel A	BRA	18.6	3.0
Panel A	ESP	12.2	2.8
Panel A	FRA	3.1	2.5
Panel A	CHL	9.6	1.4
Panel B	LIE	0.0	4.9
Panel B	CHE	0.3	3.9
Panel B	PAN	1.4	3.0
Panel B	LUX	0.1	1.0
Panel B	IRL	0.0	0.9
Panel B	CYM	0.2	0.6
Panel B	BMU	0.0	0.7
Panel B	SGP	0.0	0.5
Panel B	VGB	0.0	0.3
Panel B	BHS	0.0	0.2

Notes: Home share: percent of the parent's exports shipped to its own jurisdiction. Panel A: largest parent countries, excluding tax havens. Panel B: largest tax-haven and conduit jurisdictions. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

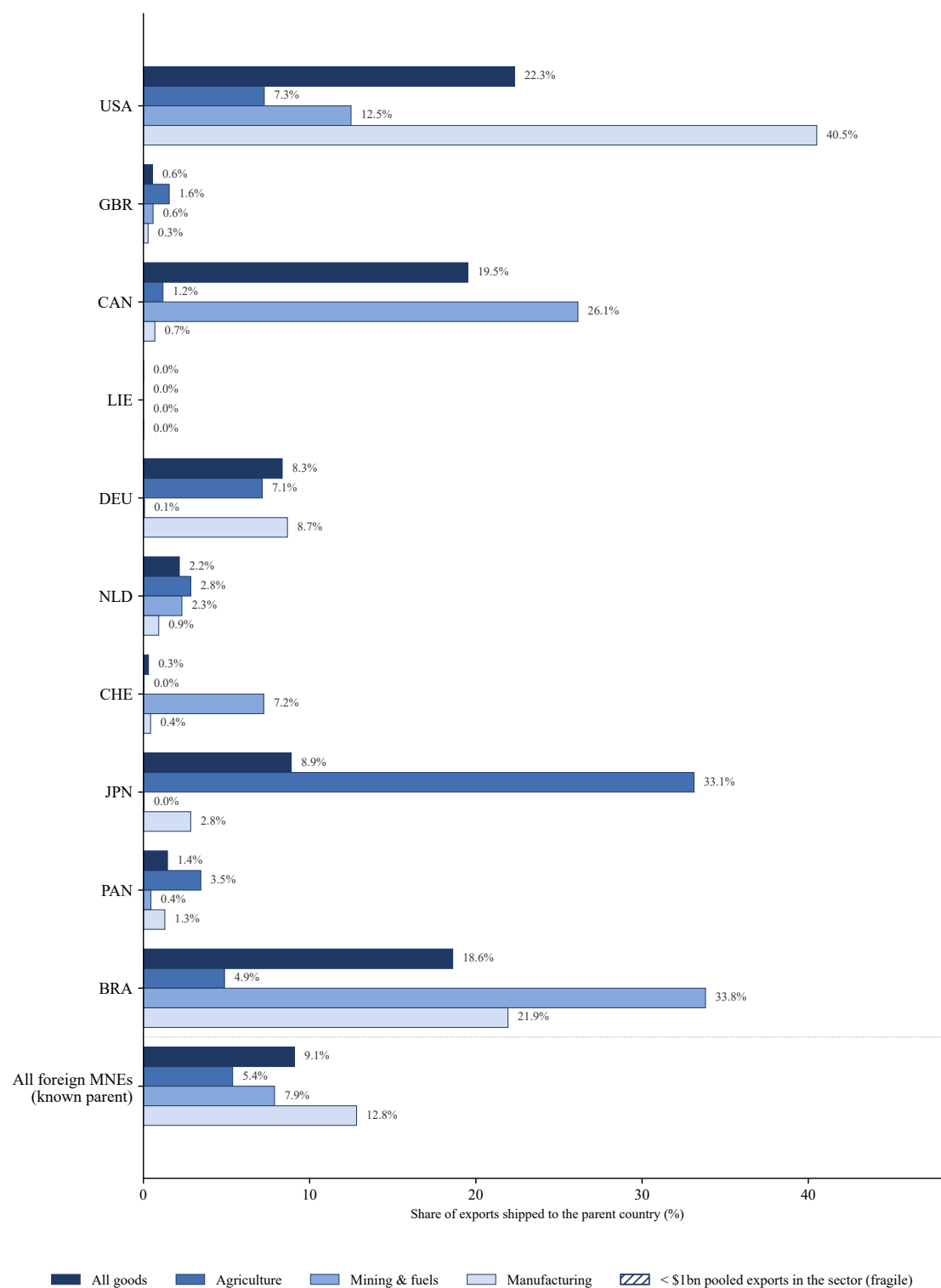


Figure 27: Share of Foreign-Multinational Exports Shipped to the Parent Country, by Sector

Notes: Percent of each parent country's exports shipped to that same country, by sector. Hatched bars: the parent country's exports in the sector are below US\$1 billion over the period.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

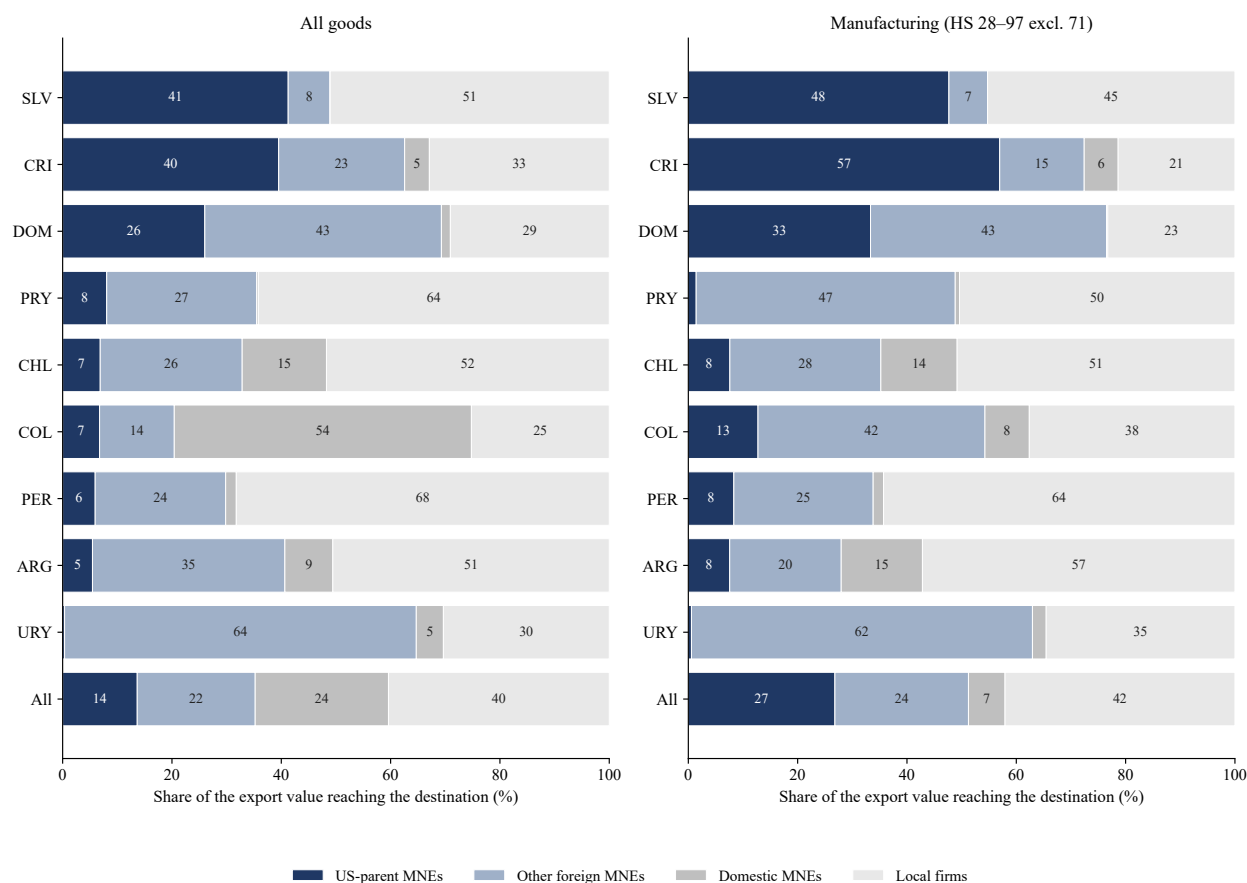
Table 27: Share of Foreign-Multinational Exports Shipped to the Parent Country, by Sector

Parent	% of the parent's exports shipped to the parent's country				Parent's exports (US\$bn/yr)			
	All goods	Agriculture	Mining & fuels	Manufacturing	All goods	Agriculture	Mining & fuels	Manufacturing
USA	22.3	7.3	12.5	40.5	23.1	6.7	7.3	9.1
GBR	0.6	1.6	0.6	0.3	16.7	0.7	11.9	4.2
CAN	19.5	1.2	26.1	0.7	7.4	0.4	5.7	1.3
LIE	0.0	0.0	0.0	0.0	4.9	2.2	2.0	0.7
DEU	8.3	7.1	0.1	8.7	4.6	0.8	0.0	3.7
NLD	2.2	2.8	2.3	0.9	4.4	2.7	0.1	1.5
CHE	0.3	0.0	7.2	0.4	3.9	3.1	0.1	0.7
JPN	8.9	33.1	0.0	2.8	3.5	0.6	0.0	2.9
PAN	1.4	3.5	0.4	1.3	3.0	0.6	1.0	1.3
BRA	18.6	4.9	33.8	21.9	3.0	1.1	0.7	1.3
All known-parent foreign MNEs	9.1	5.4	7.9	12.8	97.6	27.2	34.5	35.7

Notes: Left panel: percent of each parent country's exports shipped to that same country. Right panel: the parent country's exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

3.5 Destination Markets

**Figure 28: Exports to the United States, by Type of Exporter and Exporting Country**

Notes: Percent of each country's exports to the United States, by type of exporter. Left: all goods. Right: manufacturing.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 28: Exports to the United States, by Type of Exporter and Sector

Sector	Exports to the USA (US\$bn/yr)	US-parent MNEs (%)	Other foreign MNEs (%)	Domestic MNEs (%)	Local firms (%)
All goods	40.0	13.6	21.6	24.4	40.4
Agriculture	10.7	4.8	28.0	9.2	58.0
Mining & fuels	14.6	6.6	14.8	52.0	26.6
Manufacturing	14.5	26.8	24.4	6.7	42.1
Rest	0.1	40.0	38.4	4.5	17.1

Notes: Percent of each sector's exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

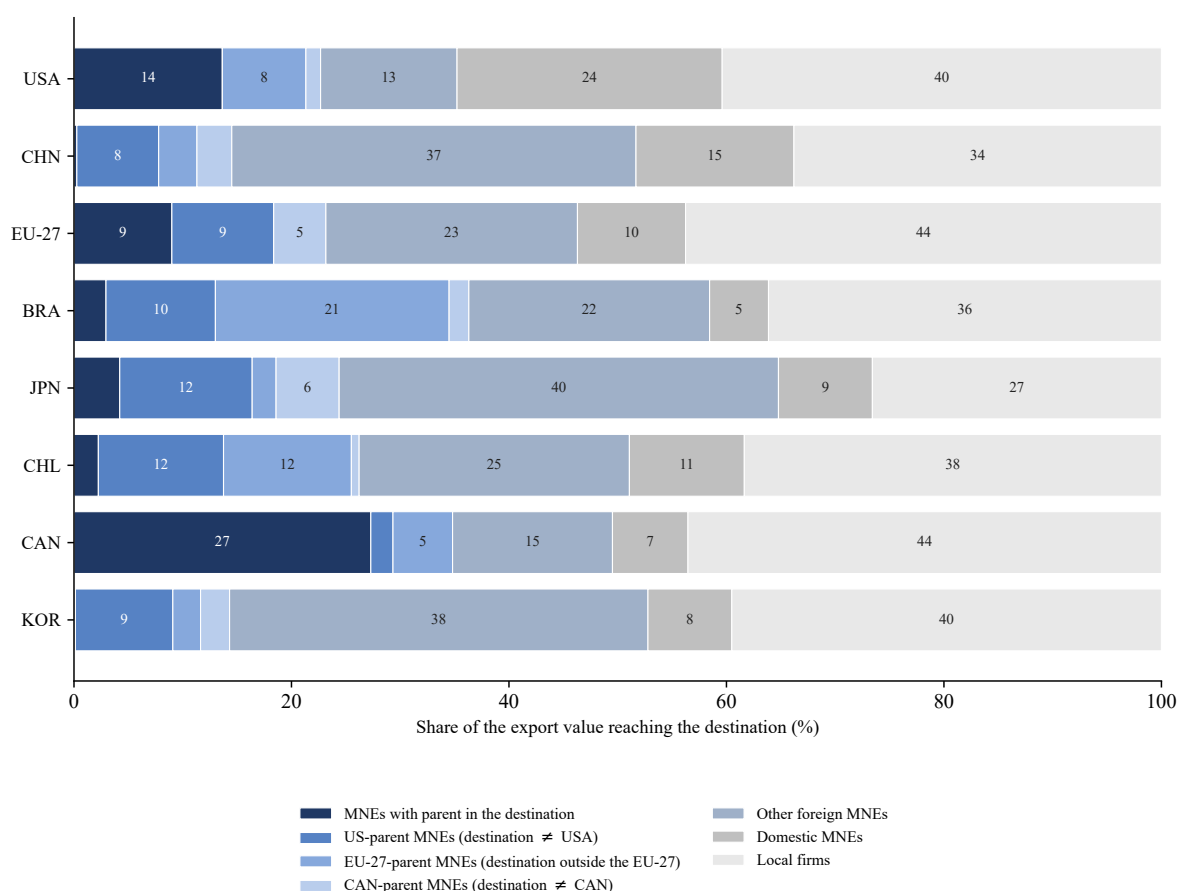
Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 29: Exports to the United States, by Type of Exporter and Exporting Country

Origin	Exports to the USA (US\$bn/yr)	US-parent MNEs (%)	Other foreign MNEs (%)	Domestic MNEs (%)	Local firms (%)
SLV	2.3	41.3	7.7	0.0	51.1
CRI	4.0	39.5	23.1	4.5	32.9
DOM	4.1	26.0	43.3	1.6	29.1
PRY	0.2	8.1	27.4	0.3	64.2
CHL	4.5	6.9	26.0	15.4	51.7
COL	13.8	6.8	13.7	54.4	25.2
PER	6.4	6.0	23.9	1.9	68.2
ARG	4.2	5.5	35.2	8.8	50.6
URY	0.4	0.4	64.3	5.0	30.3
All	40.0	13.6	21.6	24.4	40.4

Notes: Percent of each country's exports to the United States, by type of exporter. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

**Figure 29: Ownership of Exports to the Main Destination Markets**

Notes: Percent of the export value reaching each destination, by type of exporter. The first segment is exports by MNEs whose parent is located in that destination. EU-27 member states are pooled.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 30: Ownership of Exports to the Main Destination Markets

Destination	Exports (US\$bn/yr)	MNEs with parent in the destination (%)	US-parent MNEs (%)	EU-27- parent MNEs (%)	CAN-parent MNEs (%)	Other foreign MNEs (%)	Domestic MNEs (%)	Local firms (%)
USA	40.0	13.6	–	7.7	1.4	12.5	24.4	40.4
CHN	29.0	0.3	7.5	3.5	3.2	37.2	14.5	33.8
EU-27	29.7	9.0	9.4	–	4.8	23.1	9.9	43.8
BRA	20.0	2.9	10.1	21.5	1.8	22.1	5.4	36.1
JPN	6.5	4.2	12.2	2.2	5.8	40.4	8.6	26.6
CHL	6.6	2.2	11.5	11.8	0.7	24.9	10.6	38.4
CAN	6.0	27.3	2.0	5.5	–	14.7	6.9	43.5
KOR	5.0	0.1	9.0	2.5	2.7	38.5	7.7	39.5

Notes: Percent of the export value reaching each destination, by type of exporter. EU-27 member states are pooled. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 31: Main Products Shipped by US-Parent Multinationals to the United States

HS6	Description	US-parent exports to the USA (US\$bn/yr)	% of all US-parent exports to the USA	% of US-parent exports of the product going to the USA	All exports of the product to the USA (US\$bn/yr)	US-parent share of exports of the product to the USA (%)	Main origins of the US-parent shipments (% of the line)
610910	T-shirts, singlets & other vests, knitted/cro...	0.49	11.1	84	1.01	53	SLV 85%, DOM 10%, PER 5%
901839	Catheters, cannulae & the like	0.52	9.6	73	0.63	89	CRI 92%, DOM 8%, COL 0%
270900	Petroleum oils & oils obt. from bituminous mi...	0.41	9.1	24	7.84	5	COL 100%, ARG 0%
901890	Instruments & appliances used in medical/surg...	0.39	6.8	69	0.94	47	CRI 71%, DOM 29%, COL 0%
270112	Bituminous coal, whether/not pulverised but n...	0.19	4.3	9	0.39	48	COL 100%
854231	Electronic integrated circuits, processors & ...	0.21	3.9	28	0.22	95	CRI 100%, COL 0%, ARG 0%
401110	New pneumatic tyres, of rubber, of a kind use...	0.13	3.4	93	0.24	63	CHL 100%, DOM 0%, COL 0%
261390	Molybdenum ores & concentrates, other than ro...	0.15	2.8	99	0.18	79	PER 100%
611595	Other n.e.s. in 61.15, knitted/crocheted, of ...	0.10	2.4	100	0.11	97	SLV 100%, DOM 0%
610711	Men's/boys' underpants & briefs, knitted/croc...	0.09	2.2	87	0.10	87	SLV 100%, DOM 0%, CRI 0%
621210	Brassieres & parts thereof, whether/not knit...	0.09	1.8	98	0.12	78	SLV 74%, DOM 26%, CRI 0%
080430	Pineapples, fresh/dried	0.08	1.6	66	0.42	20	CRI 100%
240210	Cigars, cheroots & cigarillos containing toba...	0.09	1.3	100	0.57	16	DOM 100%
740311	Cathodes & sections of cathodes, of refined c...	0.06	1.2	24	1.56	3	PER 64%, CHL 36%
640510	Footwear with uppers of leather/composition l...	0.08	1.2	70	0.11	71	DOM 100%
All lines		5.0	100.0	22	40.0	14	

Notes: Fifteen HS6 products with the largest exports by US-parent MNEs to the United States. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

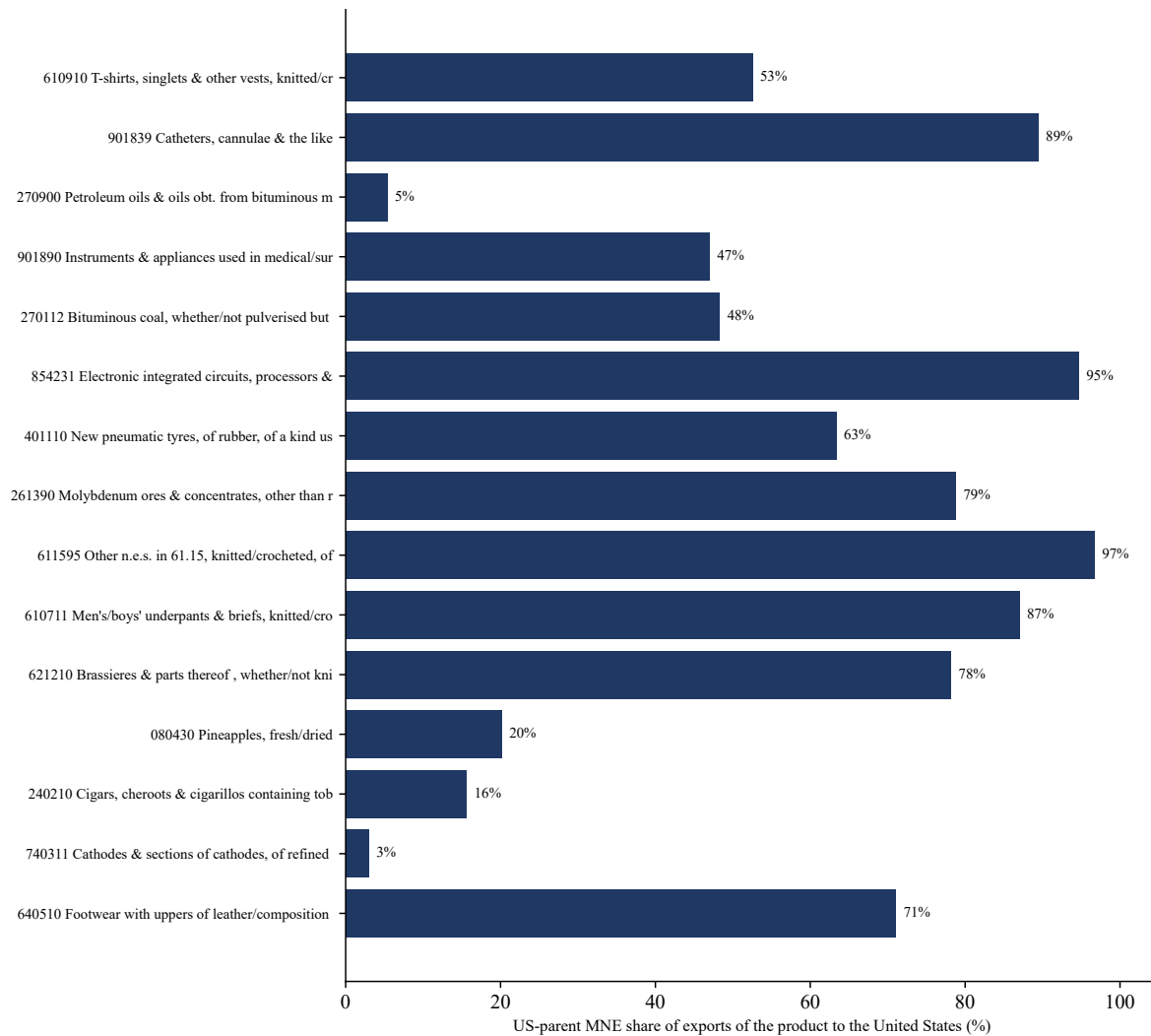


Figure 30: US-Parent Share of Exports to the United States, Main Products Shipped by US-Parent Multinationals

Notes: Percent of the nine countries' exports of each product to the United States shipped by US-parent MNEs; the fifteen products with the largest US-parent exports to the United States.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 32: Largest Export Products to the United States, by Type of Exporter

HS6	Description	Exports to the USA (US\$bn/yr)	US-parent MNEs (%)	Other foreign MNEs (%)	Domestic MNEs (%)	Local firms (%)
270900	Petroleum oils & oils obt. from bituminous mi...	7.84	5	10	81	4
710812	Gold (incl. gold plated with platinum), in un...	2.72	1	16	9	74
740311	Cathodes & sections of cathodes, of refined c...	1.56	3	33	4	60
090111	Coffee, not roasted, not decaffeinated	1.42	3	36	8	53
271019	Petroleum oils & oils obtained from bituminou...	1.14	0	14	51	35
610910	T-shirts, singlets & other vests, knitted/cro...	1.01	53	7	0	40
901890	Instruments & appliances used in medical/surg...	0.94	47	44	0	9
060319	Cut flowers & flower buds of a kind suit. for...	0.61	1	19	19	60
271011	Light petroleum oils & preparations	0.69	4	14	5	76
901839	Catheters, cannulae & the like	0.63	89	2	0	9
030441	Fish fillets&other fish meat (excl. of 0304.1...	0.38	2	18	24	56
080610	Grapes, fresh	0.39	5	27	10	58
270112	Bituminous coal, whether/not pulverised but n...	0.39	48	32	1	19
240210	Cigars, cheroots & cigarillos containing toba...	0.57	16	47	0	37
080390	Bananas, fresh or dried (excl. plantains) [HS...	0.43	14	27	1	58

Notes: Fifteen largest HS6 products exported to the United States; percent of each product's exports to the United States by type of exporter. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4 Fact 4: A Small Set of Large Multinational Groups Accounts for Most Exports

4.1 Size of the Group's Global Network

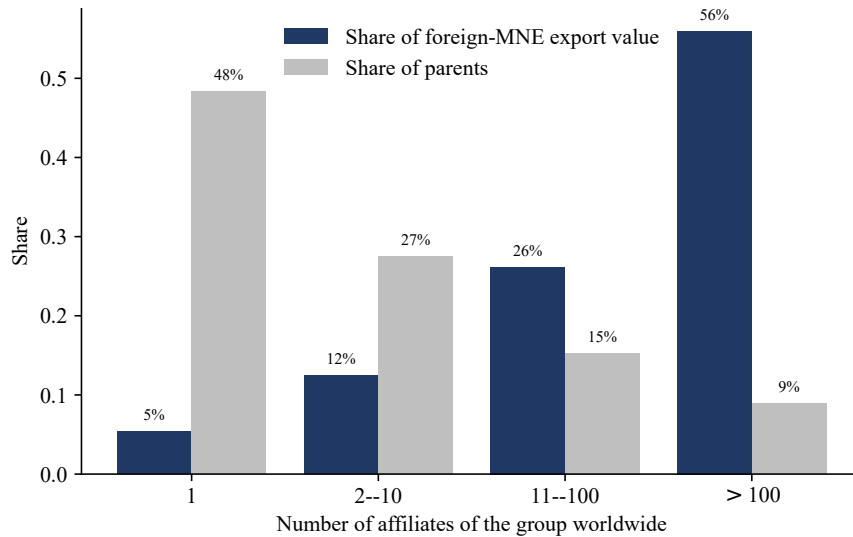


Figure 31: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

Notes: Shares of foreign-MNE export value and of parents, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 33: Foreign-Multinational Exports and Number of Parents, by Size of the Group's Global Network

Affiliates worldwide	Parents	Share of parents (%)	Share of foreign-MNE export value (%)
1	5,982	48.4	5.4
2-10	3,397	27.5	12.4
11-100	1,893	15.3	26.2
>100	1,099	8.9	56.0
All parents	12,371	100	100

Notes: Number and share of parents, and share of foreign-MNE export value, by the number of affiliates the group has worldwide.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.2 Concentration within Products

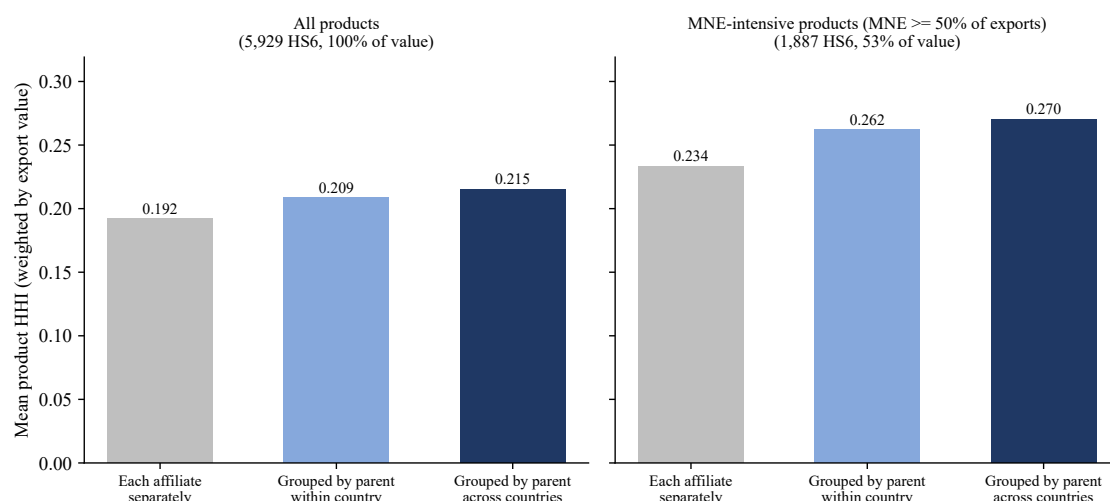


Figure 32: Export Concentration within Products: Firms and Groups

Notes: Mean Herfindahl index of exporters within each HS6 product, weighted by export value, counting each affiliate separately or grouping affiliates by parent within or across countries. Right panel: products where MNEs account for at least half of exports.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 34: Export Concentration within Products: Firms and Groups

Sample	Measure	Firm level	Groups within country	Groups across countries
<i>All products: 5,929 HS6, 100% of value</i>				
	Mean HHI	0.192	0.209	0.215
	Mean top-exporter share	31.0%	32.5%	33.7%
	Effective number of exporters (1/HHI)	12.5	11.7	10.4
<i>MNE-intensive products (MNE ≥ 50% of exports): 1,887 HS6, 53% of value</i>				
	Mean HHI	0.234	0.262	0.270
	Mean top-exporter share	36.0%	38.7%	39.8%
	Effective number of exporters (1/HHI)	8.9	8.0	7.4

Notes: Means across HS6 products, weighted by export value. Firm level: each affiliate counted separately. Groups: affiliates grouped by ultimate parent within each exporting country or across countries.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

4.3 Groups and Affiliates

Table 35: Multinational Affiliates and Groups in Origin–Destination–Product–Year Cells

Origin-destination-product-year cells with ≥ 1 MNE	
Cells	1,008,755
MNE affiliates per cell (mean)	1.91
MNE groups per cell (mean)	1.90
Affiliates per group, pooled	1.01
Affiliates per group, mean weighted by export value	1.08
Cells where one group exports through > 1 affiliate (%)	1.0

Notes: Cells with at least one MNE. Group: ultimate parent in Orbis or Dun & Bradstreet (parent name when the identifier is missing).

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 36: Multinational Presence and Exports: Affiliates and Groups

	Origin × year, destination × year, product FE			Origin × dest. × product, origin × dest. × year FE		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.0685*** (0.0368)			0.9414*** (0.0206)		
ln(# MNE groups)		2.0728*** (0.0371)	2.0617*** (0.0365)		0.9416*** (0.0208)	0.9421*** (0.0208)
ln(affiliates per group)			2.7723*** (0.2413)			0.8692*** (0.0648)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.3406*** (0.0319)			0.2389*** (0.0178)		
ln(# MNE groups)		1.3471*** (0.0319)	1.3427*** (0.0317)		0.2400*** (0.0177)	0.2400*** (0.0177)
ln(affiliates per group)			1.1055*** (0.2963)			0.0605 (0.0985)
Product FE	✓	✓	✓			
Origin × year FE	✓	✓	✓			
Destination × year FE	✓	✓	✓			
Origin × dest. × product FE				✓	✓	✓
Origin × dest. × year FE				✓	✓	✓
Observations (Panel A)	1,008,185	1,008,185	1,008,185	883,545	883,545	883,545
Observations (Panel B)	475,514	475,514	475,514	429,882	429,882	429,882

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNE affiliates or ln number of MNE groups (distinct ultimate parents). Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

5 Fact 5: Multinational Presence Is Associated with Higher Exports

Table 37: Multinational Presence and Exports

	Intensive: ln(# MNE firms)			Extensive: any MNE present		
	(1)	(2)	(3)	(1)	(2)	(3)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.0823*** (0.0363)	2.0685*** (0.0368)	0.9414*** (0.0206)			
Any MNE present				1.4622*** (0.0394)	1.4591*** (0.0388)	0.6818*** (0.0165)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.3536*** (0.0318)	1.3406*** (0.0319)	0.2389*** (0.0178)			
Any MNE present				1.4168*** (0.0371)	1.3996*** (0.0367)	0.1311*** (0.0102)
Origin FE	✓			✓		
Destination FE	✓			✓		
Product FE	✓	✓		✓	✓	
Year FE	✓			✓		
Origin × year FE		✓			✓	
Destination × year FE		✓			✓	
Origin × dest. × product FE			✓			✓
Origin × dest. × year FE			✓			✓
Observations (Panel A)	1,008,472	1,008,185	883,545	1,829,852	1,829,626	1,633,319
Observations (Panel B)	475,856	475,514	429,882	1,297,322	1,297,111	1,137,550

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Intensive margin: ln number of MNEs, cells with at least one MNE. Extensive margin: indicator for at least one MNE. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 38: Multinational Exports by Exporting Country and Destination Type

Origin	MNE value (US\$bn/yr)	of which domestic (%)	Foreign-MNE value (US\$bn/yr)	% of foreign-MNE value, by destination	
				parent country	other (of which: group affiliate there)
ARG	37.5	13.3	32.5	6.5	93.5 (41.8)
CHL	21.0	22.1	16.4	4.3	95.7 (56.7)
COL	32.8	52.5	15.5	6.5	93.5 (34.2)
CRI	6.6	5.7	6.3	25.5	74.5 (33.8)
DOM	6.0	1.8	5.9	29.0	71.0 (21.6)
PER	19.3	5.0	18.3	3.9	96.1 (28.8)
PRY	4.5	1.8	4.4	2.1	97.9 (65.5)
SLV	2.4	0.6	2.4	40.7	59.3 (28.1)
URY	4.4	15.9	3.7	5.2	94.8 (37.7)
All	134.5	23.4	105.4	8.4	91.6 (40.5)

Notes: MNE and foreign-MNE export value by country, and the split of foreign-MNE value by destination. Parent country: exports to the country of the ultimate parent. Other: all remaining destinations; in parentheses, destinations where the group has another affiliate. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 39: Multinational Presence and Exports: Intensive Margin, by Type of Multinational

	All MNEs		Foreign / domestic		Foreign by destination, domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
ln(# MNE firms)	2.0685*** (0.0368)	0.9414*** (0.0206)				
ln(# foreign MNEs)			1.2129*** (0.0285)	0.4506*** (0.0124)		
ln(# foreign MNEs, parent-country destination)					0.3970*** (0.0856)	0.1448*** (0.0370)
ln(# foreign MNEs, other destinations)					0.9465*** (0.0536)	0.3133*** (0.0382)
ln(# domestic MNEs)			0.4382*** (0.0259)	0.2214*** (0.0151)	0.3770*** (0.0514)	0.1555*** (0.0315)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
ln(# MNE firms)	1.3406*** (0.0319)	0.2389*** (0.0178)				
ln(# foreign MNEs)			1.1331*** (0.0375)	0.2413*** (0.0217)		
ln(# foreign MNEs, parent-country destination)					0.2934*** (0.0882)	0.1945*** (0.0527)
ln(# foreign MNEs, other destinations)					1.0330*** (0.0607)	0.1951*** (0.0648)
ln(# domestic MNEs)			0.4707*** (0.0341)	0.1250*** (0.0218)	0.4142*** (0.0633)	0.0946* (0.0552)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (Panel A)	1,008,185	883,545	113,422	98,223	9,197	7,524
Observations (Panel B)	475,514	429,882	93,010	81,563	8,780	7,255

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: ln number of MNEs of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent's country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 40: Multinational Presence and Exports: Extensive Margin, by Type of Multinational

	All MNEs		Foreign / domestic		Foreign by destination, domestic	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Dependent variable: ln exports, all firms</i>						
Any MNE present	1.4591*** (0.0388)	0.6818*** (0.0165)				
Any foreign MNE			1.4990*** (0.0339)	0.6986*** (0.0158)		
Any foreign MNE, parent-country destination					1.3873*** (0.0951)	0.5909*** (0.0431)
Any foreign MNE, other destination					1.5093*** (0.0319)	0.6976*** (0.0146)
Any domestic MNE			1.4670*** (0.0416)	0.4917*** (0.0165)	1.4334*** (0.0388)	0.4909*** (0.0166)
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>						
Any MNE present	1.3996*** (0.0367)	0.1311*** (0.0102)				
Any foreign MNE			1.2983*** (0.0346)	0.1356*** (0.0106)		
Any foreign MNE, parent-country destination					1.0183*** (0.0900)	0.1265*** (0.0194)
Any foreign MNE, other destination					1.2782*** (0.0326)	0.1386*** (0.0107)
Any domestic MNE			1.1316*** (0.0428)	0.1370*** (0.0138)	1.0959*** (0.0403)	0.1367*** (0.0138)
Product FE	✓		✓		✓	
Origin × year FE	✓		✓		✓	
Destination × year FE	✓		✓		✓	
Origin × dest. × product FE		✓		✓		✓
Origin × dest. × year FE		✓		✓		✓
Observations (Panel A)	1,829,626	1,633,319	1,829,626	1,633,319	1,829,626	1,633,319
Observations (Panel B)	1,297,111	1,137,550	1,297,111	1,137,550	1,297,111	1,137,550

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Regressors: indicators for at least one MNE of each type. Columns (5)–(6) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 41: Multinational Presence and Exports by Sector: Intensive Margin

Sample	All goods	Agriculture	Mining & fuels	Manufacturing
<i>Panel A. Dependent variable: ln exports, all firms</i>				
Fixed effects: origin $\times$ year, destination $\times$ year, product				
ln(# foreign MNEs)	1.2129*** (0.0285)	1.2954*** (0.0275)	1.1301*** (0.1799)	1.1644*** (0.0358)
ln(# domestic MNEs)	0.4382*** (0.0259)	0.5233*** (0.0266)	0.3172 (0.2062)	0.3833*** (0.0319)
Observations	113,422	25,437	1,417	85,453
Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year				
ln(# foreign MNEs)	0.4506*** (0.0124)	0.4625*** (0.0188)	0.2479 (0.1778)	0.4435*** (0.0148)
ln(# domestic MNEs)	0.2214*** (0.0151)	0.2831*** (0.0234)	0.3216* (0.1639)	0.1904*** (0.0175)
Observations	98,223	22,002	1,027	73,318
<i>Panel B. Dependent variable: ln exports, non-MNE firms</i>				
Fixed effects: origin $\times$ year, destination $\times$ year, product				
ln(# foreign MNEs)	1.1331*** (0.0375)	1.2191*** (0.0363)	0.9420*** (0.2641)	1.0546*** (0.0493)
ln(# domestic MNEs)	0.4707*** (0.0341)	0.5676*** (0.0339)	-0.6594 (0.4830)	0.4194*** (0.0419)
Observations	93,010	22,503	995	68,534
Fixed effects: origin $\times$ destination $\times$ product, origin $\times$ destination $\times$ year				
ln(# foreign MNEs)	0.2413*** (0.0217)	0.3602*** (0.0292)	-0.1512 (0.3320)	0.1865*** (0.0250)
ln(# domestic MNEs)	0.1250*** (0.0218)	0.2461*** (0.0341)	-0.7273 (0.6536)	0.0742*** (0.0238)
Observations	81,563	19,535	726	59,772

Notes: Observations are origin–destination–HS6–year cells. Dependent variable: ln exports of the cell (Panel A) and of the non-MNE firms in the cell (Panel B). Specification of columns (3)–(4) of Table 39, estimated by sector. Blank: too few observations. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

6 Fact 6: Distance Is a Weaker Barrier for Multinationals

Table 42: Distance and Firm Exports: Multinationals by Presence at the Destination

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1639*** (0.0454)	-0.1848*** (0.0460)	-0.1632*** (0.0452)	-0.1838*** (0.0459)
ln distance $\times$ MNE	0.0461*** (0.0038)	0.0458*** (0.0038)		
ln distance $\times$ MNE, present at destination			0.0689*** (0.0070)	0.0682*** (0.0070)
ln distance $\times$ MNE, not present at destination			0.0393*** (0.0032)	0.0391*** (0.0032)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin $\times$ year FE		✓		✓
Destination $\times$ year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Non-MNE firms are the omitted group. Present: the group has an affiliate in the destination or the destination is the parent's country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 43: Distance and Firm Exports: Foreign and Domestic Multinationals

Dep. var.: ln exports	(1)	(2)	(3)	(4)
ln distance	-0.1645*** (0.0454)	-0.1853*** (0.0460)	-0.1644*** (0.0454)	-0.1851*** (0.0460)
ln distance × foreign MNE	0.0487*** (0.0041)	0.0483*** (0.0040)		
ln distance × foreign MNE, parent-country destination			0.0642*** (0.0159)	0.0638*** (0.0161)
ln distance × foreign MNE, other destination			0.0477*** (0.0035)	0.0473*** (0.0035)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0318*** (0.0042)	0.0317*** (0.0042)	0.0317*** (0.0042)
Origin FE	✓		✓	
Destination FE	✓		✓	
Year FE	✓		✓	
Origin × year FE		✓		✓
Destination × year FE		✓		✓
Product FE	✓	✓	✓	✓
Observations	5,081,547	5,081,360	5,081,547	5,081,360

Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Non-MNE firms are the omitted group. Columns (3)–(4) split foreign MNEs by whether the destination is the parent’s country. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 44: Distance and Firm Exports, by Sector

Dep. var.: ln firm exports	All goods	Agriculture	Mining & fuels	Manufacturing
Origin, destination, year and product FE				
ln distance	-0.1645*** (0.0454)	-0.2135*** (0.0748)	-0.0926 (0.0819)	-0.1510*** (0.0454)
ln distance × foreign MNE	0.0487*** (0.0041)	0.0366*** (0.0034)	0.0566*** (0.0086)	0.0515*** (0.0048)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0187*** (0.0046)	0.0706*** (0.0180)	0.0388*** (0.0052)
Observations	5,081,547	1,045,009	79,938	3,917,960
Origin × year, destination × year and product FE				
ln distance	-0.1853*** (0.0460)	-0.2285*** (0.0711)	-0.1001 (0.0828)	-0.1742*** (0.0459)
ln distance × foreign MNE	0.0483*** (0.0040)	0.0358*** (0.0034)	0.0557*** (0.0087)	0.0512*** (0.0048)
ln distance × domestic MNE	0.0318*** (0.0042)	0.0182*** (0.0046)	0.0673*** (0.0183)	0.0390*** (0.0052)
Observations	5,081,360	1,044,787	79,612	3,917,765

Notes: Observations are firm–destination–HS6–year cells. Dependent variable: ln exports. Specification of columns (1)–(2) of Table 43, estimated by sector. Standard errors clustered by origin–destination pair in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors’ calculations based on customs data matched to Orbis and Dun & Bradstreet.

7 Products

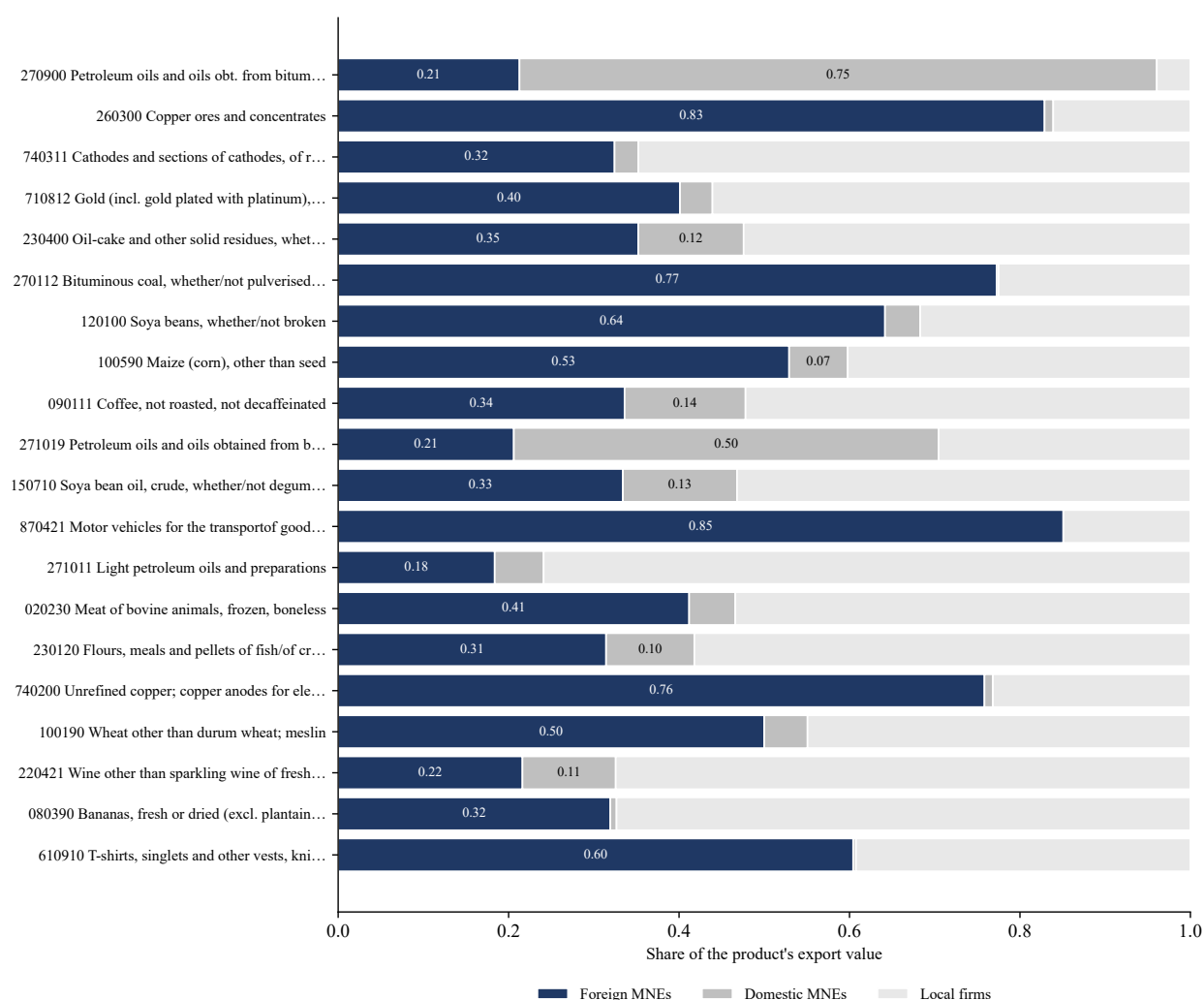


Figure 33: Ownership of the Twenty Largest Export Products

Notes: Shares of each product's export value. Twenty largest HS6 products by export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

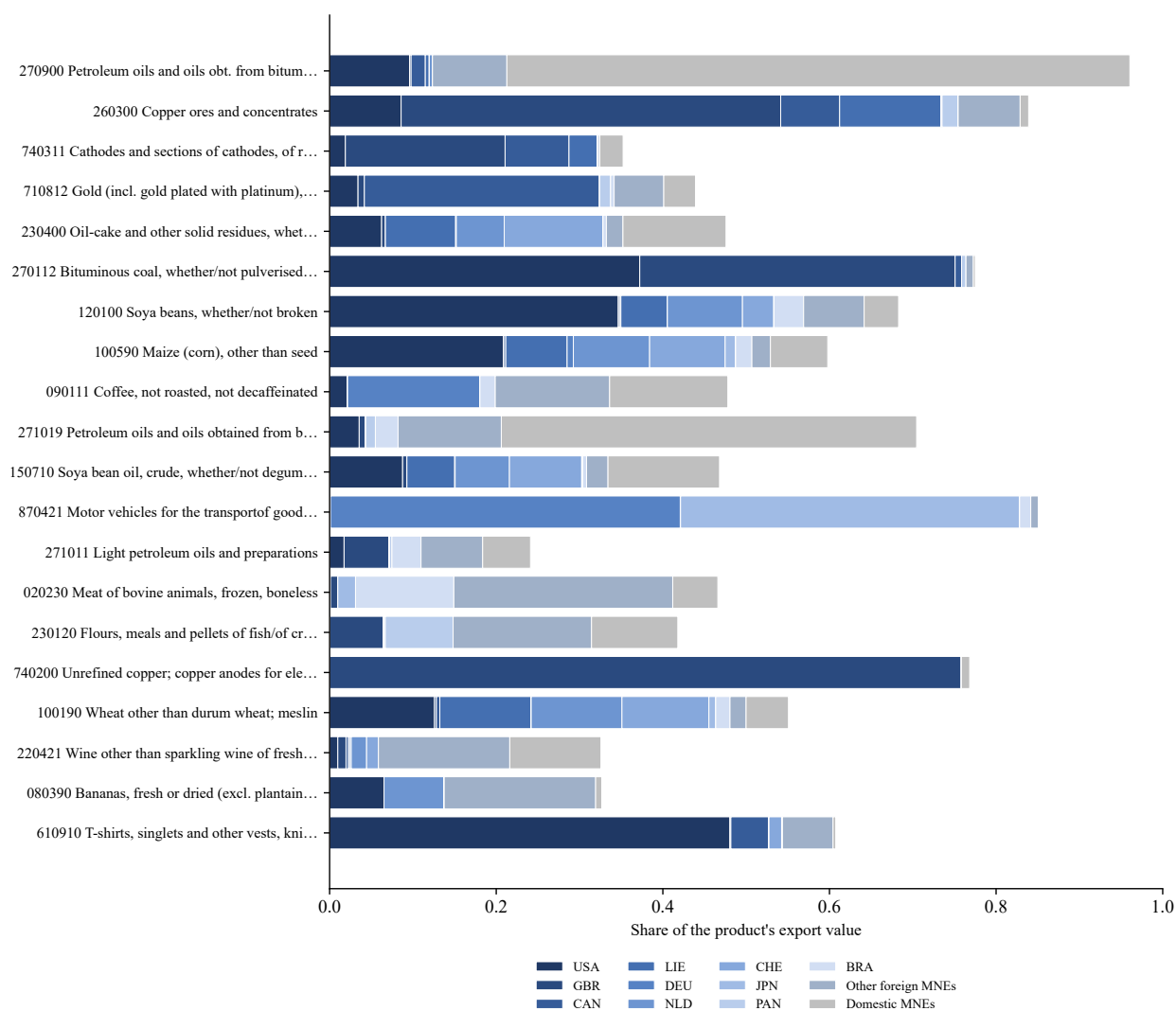


Figure 34: Ownership of the Twenty Largest Export Products, by Parent Country

Notes: Shares of each product's export value. Foreign-MNE share split by country of the ultimate parent; the unfilled part of each bar is local firms.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

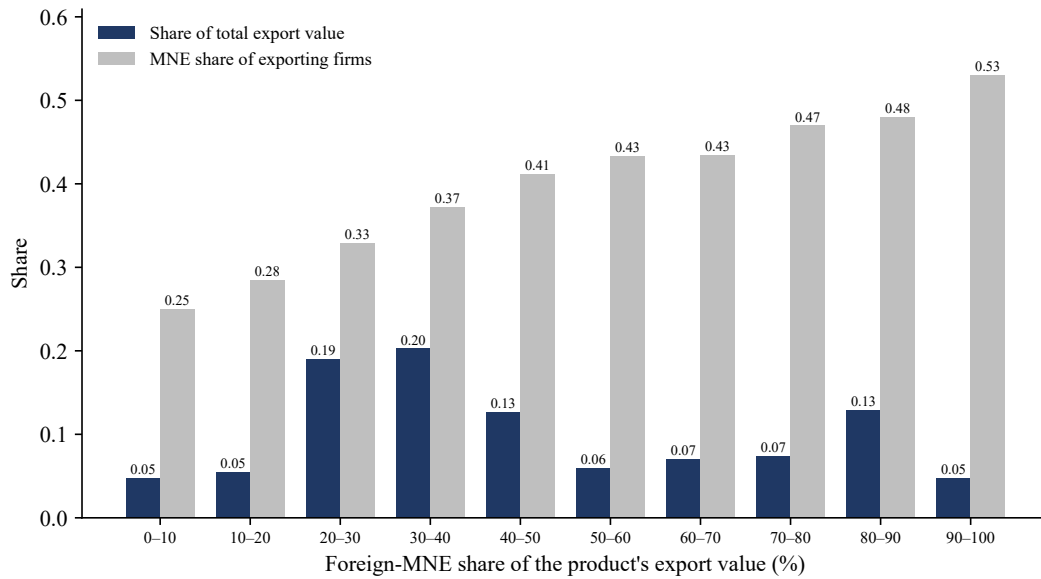


Figure 35: Distribution of Export Value by Product-Level Foreign-Multinational Share

Notes: Share of export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 45: Distribution of Export Value by Product-Level Foreign-Multinational Share

Foreign share bin (%)	Share of value	MNE share of exporting firms	HS6 products
0-10	0.048	0.250	1,426
10-20	0.054	0.285	731
20-30	0.189	0.329	652
30-40	0.202	0.372	591
40-50	0.127	0.411	519
50-60	0.059	0.433	442
60-70	0.070	0.435	386
70-80	0.074	0.470	311
80-90	0.129	0.479	311
90-100	0.048	0.530	560

Notes: Share of export value and MNE share of exporting firms, by the foreign-MNE share of each HS6 product's export value. MNE share of exporting firms: MNE firm-destination-product-year observations over all observations in the bin.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

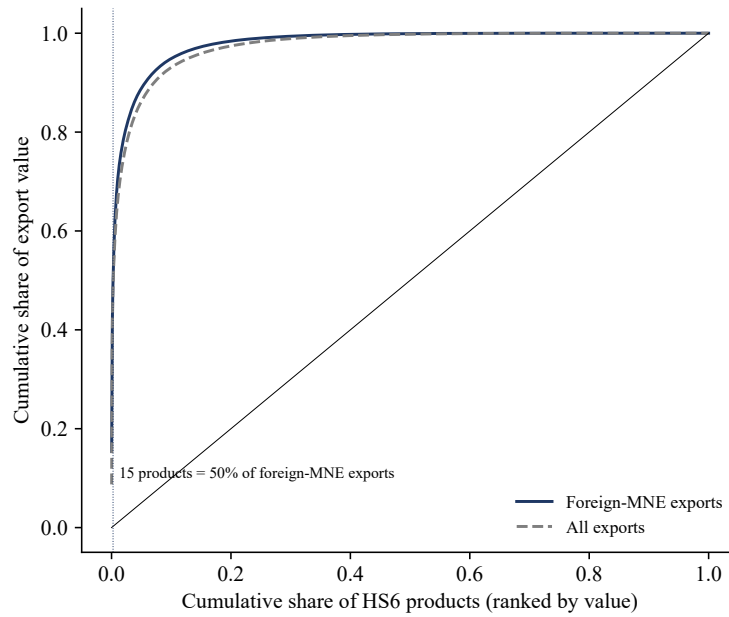


Figure 36: Concentration of Exports across Products

Notes: Cumulative share of export value against the cumulative share of HS6 products, ranked by value, for foreign-MNE exports and for all exports.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 46: Exports and Multinational Shares, by HS Section

HS section	US\$b/yr	Foreign (%)	Domestic (%)	Local (%)	Leading parent
V Mineral products	57.1	51	29	19	GBR (0.42)
II Vegetable products	33.5	42	7	51	USA (0.33)
IV Prepared foodstuffs, beverages, tobacco	25.9	37	10	53	USA (0.21)
XV Base metals	19.6	41	3	56	GBR (0.44)
XIV Precious metals and stones	15.3	41	4	56	CAN (0.65)
I Live animals; animal products	12.5	34	14	51	SAU (0.19)
VI Chemicals	11.6	49	14	37	USA (0.25)
XVII Transport equipment	9.7	78	4	18	DEU (0.30)
XI Textiles and apparel	6.9	49	3	48	USA (0.62)
III Fats and oils	6.9	31	10	59	CHE (0.26)
XVI Machinery and electrical equipment	6.2	63	4	33	USA (0.41)
VII Plastics and rubber	5.8	51	8	42	USA (0.34)
X Pulp, paper	3.4	54	28	18	PAN (0.54)
XVIII Optical, precision instruments	3.4	72	8	19	USA (0.66)
IX Wood and articles	1.9	27	34	39	BHS (0.24)
VIII Hides, skins, leather	1.6	40	12	49	BGR (0.19)
XIII Stone, ceramics, glass	0.9	49	4	46	USA (0.32)
XX Miscellaneous manufactures	0.9	53	7	40	USA (0.48)
Unclassified	0.5	49	29	23	USA (0.39)
XII Footwear, headgear	0.5	48	1	50	USA (0.66)
XXI Works of art	0.1	58	1	41	BGR (0.94)
XIX Arms	0.0	1	0	99	CHE (0.82)
Special / unclassified (98-99)	0.0	13	1	87	USA (0.57)
All sections	224.5	46	14	40	

Notes: Foreign, domestic and local: percent of each section's export value. Leading parent: largest parent country and its share of the section's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 47: Thirty Largest Export Products

HS6	Description	US\$bn/yr	Foreign (%)	Domestic (%)	Local (%)	Leading parent
270900	Petroleum oils and oils obt. from bituminous mins., crude	17.9	21	75	4	USA (0.45)
260300	Copper ores and concentrates	17.6	83	1	16	GBR (0.55)
740311	Cathodes and sections of cathodes, of refined copper, unwrought	10.4	32	3	65	GBR (0.59)
710812	Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)	13.4	40	4	56	CAN (0.70)
230400	Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...	10.8	35	12	52	CHE (0.34)
270112	Bituminous coal, whether/not pulverised but not agglomerated	5.7	77	0	23	GBR (0.49)
120100	Soya beans, whether/not broken	6.2	64	4	32	USA (0.54)
100590	Maize (corn), other than seed	4.8	53	7	40	USA (0.40)
090111	Coffee, not roasted, not decaffeinated	3.7	34	14	52	DEU (0.47)
271019	Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...	3.6	21	50	30	ESP (0.52)
150710	Soya bean oil, crude, whether/not degummed, not chemically modified	4.3	33	13	53	USA (0.26)
870421	Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...	3.2	85	0	15	DEU (0.49)
271011	Light petroleum oils and preparations	2.2	18	6	76	GBR (0.29)
020230	Meat of bovine animals, frozen, boneless	2.1	41	5	53	SAU (0.53)
230120	Flours, meals and pellets of fish/of crustaceans, molluscs/other aquatic invertebrates	1.7	31	10	58	BMU (0.38)
740200	Unrefined copper; copper anodes for electrolytic refining	1.3	76	1	23	GBR (1.00)
100190	Wheat other than durum wheat; meslin	1.9	50	5	45	USA (0.25)
220421	Wine other than sparkling wine of fresh grapes, incl. fortified; grape must with fermentation ...	1.4	22	11	67	FRA (0.25)
080390	Bananas, fresh or dried (excl. plantains) [HS2012 code]	1.4	32	1	67	JOR (0.38)
610910	T-shirts, singlets and other vests, knitted/crocheted, of cotton	1.3	60	0	39	USA (0.79)
260800	Zinc ores and concentrates	1.5	72	0	27	GBR (0.67)
080610	Grapes, fresh	1.1	29	9	62	ESP (0.29)
260111	Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated	1.1	2	47	51	CAN (0.65)
260700	Lead ores and concentrates	1.3	58	0	42	GBR (0.69)
870323	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	1.3	75	0	25	FRA (0.45)
020130	Meat of bovine animals, fresh/chilled, boneless	1.4	35	11	55	SAU (0.63)
901890	Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electro-m...	1.4	89	0	11	USA (0.52)
300490	Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...	1.0	55	7	38	USA (0.29)
382490	Chemical products and preparations of the chemical/allied industries (including those consisting...	1.2	36	14	50	LIE (0.61)
470321	Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, conif...	0.7	71	29	0	PAN (0.82)

Notes: Thirty largest HS6 products by export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 48: Thirty Largest Export Products of Foreign Multinationals

HS6	Description	US\$bn/yr	Foreign (%)	Domestic (%)	Local (%)	Leading parent
260300	Copper ores and concentrates	17.6	83	1	16	GBR (0.55)
710812	Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)	13.4	40	4	56	CAN (0.70)
270112	Bituminous coal, whether/not pulverised but not agglomerated	5.7	77	0	23	GBR (0.49)
740311	Cathodes and sections of cathodes, of refined copper, unwrought	10.4	32	3	65	GBR (0.59)
270900	Petroleum oils and oils obt. from bituminous mins., crude	17.9	21	75	4	USA (0.45)
120100	Soya beans, whether/not broken	6.2	64	4	32	USA (0.54)
230400	Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...	10.8	35	12	52	CHE (0.34)
870421	Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...	3.2	85	0	15	DEU (0.49)
100590	Maize (corn), other than seed	4.8	53	7	40	USA (0.40)
090111	Coffee, not roasted, not decaffeinated	3.7	34	14	52	DEU (0.47)
740200	Unrefined copper; copper anodes for electrolytic refining	1.3	76	1	23	GBR (1.00)
150710	Soya bean oil, crude, whether/not degummed, not chemically modified	4.3	33	13	53	USA (0.26)
901890	Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electro-m...	1.4	89	0	11	USA (0.52)
260800	Zinc ores and concentrates	1.5	72	0	27	GBR (0.67)
870323	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	1.3	75	0	25	FRA (0.45)
610910	T-shirts, singlets and other vests, knitted/crocheted, of cotton	1.3	60	0	39	USA (0.79)
271111	Natural gas, liquefied	0.9	100	0	0	USA (1.00)
100190	Wheat other than durum wheat; meslin	1.9	50	5	45	USA (0.25)
720260	Ferro-nickel, in granular/powder form	0.8	100	0	0	ZAF (0.85)
271019	Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...	3.6	21	50	30	ESP (0.52)
020230	Meat of bovine animals, frozen, boneless	2.1	41	5	53	SAU (0.53)
901839	Catheters, cannulae and the like	0.9	89	0	11	USA (0.96)
854231	Electronic integrated circuits, processors and controllers, whether/not combined with memorie...	0.8	99	0	1	USA (0.98)
260700	Lead ores and concentrates	1.3	58	0	42	GBR (0.69)
870322	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	0.8	100	0	0	USA (0.59)
470321	Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, conif...	0.7	71	29	0	PAN (0.82)
210690	Food preparations, n.e.s.	0.7	81	1	18	USA (0.79)
300490	Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...	1.0	55	7	38	USA (0.29)
870840	Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.	0.6	99	0	1	DEU (0.74)
261610	Silver ores and concentrates	0.8	76	0	24	GBR (0.47)

Notes: Thirty largest HS6 products by foreign-MNE export value. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 49: Products with the Highest Foreign-Multinational Share

HS6	Description	US\$bn/yr	Foreign (%)	Domestic (%)	Local (%)	Leading parent
720260	Ferro-nickel, in granular/powder form	0.76	100	0	0	ZAF (0.85)
790500	Zinc plates, sheets, strip and foil	0.05	100	0	0	GBR (0.94)
790112	Zinc, not alloyed, unwrought, containing by weight <99.99% of zinc	0.22	100	0	0	BRA (1.00)
870321	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	0.53	100	0	0	NLD (0.49)
390410	Poly(vinyl chloride), not mixed with any other substance, in primary forms	0.38	100	0	0	MEX (0.83)
271111	Natural gas, liquefied	0.85	100	0	0	USA (1.00)
870322	Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...	0.82	100	0	0	USA (0.59)
870333	Vehicles principally designed for the transport of persons (excl. of 87.02 and 8703.10-8703.24)...	0.39	100	0	0	JPN (1.00)
251010	Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, unground	0.28	100	0	0	BRA (1.00)
280300	Carbon (carbon blacks and other forms of carbon, n.e.s.)	0.08	100	0	0	USA (0.99)
470311	Chemical wood pulp, soda/sulphate, other than dissolving grades, unbleached, coniferous	0.14	100	0	0	PAN (1.00)
300691	Appliances identifiable for ostomy use	0.19	100	0	0	GBR (0.80)
853620	Automatic circuit breakers, for a voltage not >1000V	0.45	99	0	1	IRL (0.89)
854231	Electronic integrated circuits, processors and controllers, whether/not combined with memorie...	0.76	99	0	1	USA (0.98)
381512	Supported catalysts, with precious metal/precious metal comps. as the active substance	0.08	99	0	1	GBR (0.80)
870840	Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.	0.63	99	0	1	DEU (0.74)
620711	Men's/boys' underpants and briefs (excl. knitted/crocheted), of cotton	0.05	98	0	2	USA (1.00)
854430	Ignition wiring sets and other wiring sets of a kind used in vehicles/aircraft/ships	0.23	98	0	2	JPN (0.81)
740710	Bars, rods and profiles, of refined copper	0.06	98	0	1	MEX (0.73)
390120	Polyethylene having a sp.gr. of 0.94/more, in primary forms	0.13	98	0	2	USA (0.98)
961900	Sanitary towels and tampons, napkins and napkin liners for babies and similar sanitary article...	0.22	97	1	1	USA (0.79)
350300	Gelatin, incl. gelatin in rectangular (incl. square) sheets, whether/not surface-worked /colou...	0.08	97	0	3	BEL (0.43)
790111	Zinc, not alloyed, unwrought, containing by weight 99.99%/more of zinc	0.35	97	0	3	BRA (0.90)
521213	Woven fabrics of cotton (excl. of 52.08-52.11), dyed, weighing not >200 g/m2	0.13	96	0	4	USA (1.00)
390190	Polymers of ethylene, in primary forms (excl. of 3901.10-3901.30)	0.13	96	0	4	USA (1.00)
840820	Compression-ignition internal combustion piston engines (diesel/semi-diesel engines) of a kind...	0.19	96	0	4	USA (0.54)
761010	Doors, windows and their frames and thresholds for doors, of aluminium	0.13	96	0	4	CYM (0.99)
870422	Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...	0.08	96	0	4	JPN (0.88)
281700	Zinc oxide; zinc peroxide	0.07	95	0	5	BRA (0.80)
300610	Sterile surgical catgut, similar sterile suture materials (including sterile absorbable surgic...	0.14	95	3	2	IRL (0.98)

Notes: HS6 products with exports of at least US\$500 million over the period, ranked by foreign-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 50: Products with the Highest Domestic-Multinational Share

HS6	Description	US\$bn/yr	Foreign (%)	Domestic (%)	Local (%)	Leading parent
282520	Lithium oxide and hydroxide	0.0	0	100	0	–
481092	Paper and paperboard(excl. of 4810.13-4810.39), coated on one/both sides with kaolin (China cl...	0.2	1	99	0	IRL (0.84)
310420	Potassium chloride	0.1	3	95	2	DZA (0.58)
441193	Fibreboard of wood/other ligneous materials, whether/not bonded with resins/other organic subs...	0.0	1	94	5	ESP (0.94)
260112	Iron ores and concentrates (excl. roasted iron pyrites), agglomerated	0.1	7	88	4	CHE (1.00)
350110	Casein	0.1	0	84	16	CHL (0.80)
020727	Cuts and edible offal of turkey, frozen	0.0	1	82	17	JPN (0.45)
730640	Tubes, pipes and hollow profiles (excl. of 7306.10-7306.30), welded, of circular cross-section...	0.1	15	82	3	DEU (0.60)
270900	Petroleum oils and oils obt. from bituminous mins., crude	17.9	21	75	4	USA (0.45)
390390	Polymers of styrene, in primary forms (excl. of 3903.11-3903.30)	0.0	15	73	12	NLD (0.78)
440711	Wood sawn/chipped length wise, sliced/peeled, whether/not planed, sanded/end-jointed, of a thi...	0.2	10	71	19	BHS (0.67)
441239	Plywood, consisting solely of sheets of wood (other than bamboo,excl. of 4412.31 and4412.32), ...	0.2	28	71	1	VGB (0.46)
390319	Polystyrene other than expansible, in primary forms	0.1	22	67	11	PAN (0.99)
040610	Fresh (unripened/uncured) cheese, incl. whey cheese, and curd	0.1	12	66	22	MUS (0.36)
283691	Lithium carbonates	0.6	14	65	21	USA (0.54)
540219	High tenacity yarn other than textured yarn/sewing thread, of nylon/other polyamides (excl. of...	0.1	36	64	0	GBR (0.80)
481141	Gummed/adhesive paper,paperboard,coated, impregnated, covered, surface-coloured, surface-decor...	0.1	26	62	12	GBR (0.56)
902139	Artificial parts of the body other than teeth, dental fittings and joints	0.4	10	60	30	VGB (0.46)
722830	Bars and rods of other alloy steel (excl. of 72.27, 7228.10 and 7228.20), not further worked t...	0.1	25	56	19	SGP (0.59)
020329	Meat of swine (excl. carcasses/half-carcasses/hams/shoulders and cuts thereof), froxen	0.2	29	54	17	JPN (1.00)
470329	Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, non-c...	0.6	46	54	0	PAN (1.00)
491110	Trade advertising material, commercial catalogue-sandthe like	0.1	16	53	31	USA (0.45)
880240	Aeroplanes and other aircraft, of an unladen weight >15000kg	0.6	40	53	7	CHL (0.99)
410449	Tanned/crust hides and skins of bovine (incl. buffalo)/equine animals, without hair on, in the...	0.1	37	52	12	URY (0.71)
271019	Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...	3.6	21	50	30	ESP (0.52)
210111	Extracts, essences and concentrates of coffee	0.2	23	48	29	CHE (0.70)
260111	Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated	1.1	2	47	51	CAN (0.65)
330300	Perfumes and toilet waters	0.1	34	46	20	PAN (0.38)
030319	Pacific salmon (<i>Oncorhynchus gorbuscha/keta/tschawytscha/kisutch/masou/rhodurus</i>), frozen (excl...	0.1	8	44	47	CAN (0.39)
391990	Self-adhesive plates, sheets, film, foil, tape, strip and other flat shapes, of plastics, whet...	0.1	41	44	15	GBR (0.57)

Notes: HS6 products with exports of at least US\$500 million over the period, ranked by domestic-MNE share. Foreign, domestic and local: percent of the product's export value. Leading parent: largest parent country and its share of the product's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

8 Sectors

Table 51: Exports and Multinational Shares, by Sector

Sector	Total exports (US\$bn/yr)	MNE exports (US\$bn/yr)	Foreign (%)	Domestic (%)	Leading parents (share of foreign)
Agriculture	78.9	37.8	38.1	9.5	USA 0.24, CHE 0.11, NLD 0.10
Mining & fuels	72.4	52.4	49.2	24.5	GBR 0.36, USA 0.21, CAN 0.16
Manufacturing	72.7	43.9	51.6	8.0	USA 0.25, GBR 0.14, DEU 0.09
Rest	0.5	0.4	47.9	28.0	USA 0.39, GBR 0.12, ESP 0.11
All	224.5	134.5	46.3	14.1	

Notes: Foreign and domestic: percent of the sector's export value. Other goods: HS 98–99 and unclassified codes. Leading parents: the three largest parent countries and their share of the row's foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

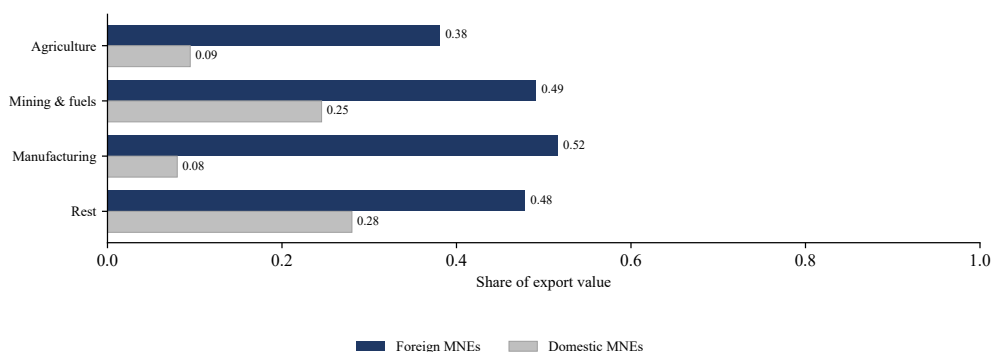


Figure 37: Multinational Share of Export Value, by Sector

Notes: Shares of export value, all available years pooled. Agriculture: HS 01–24. Mining and fuels: HS 25–27 and 71. Manufacturing: HS 28–97 excluding 71.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

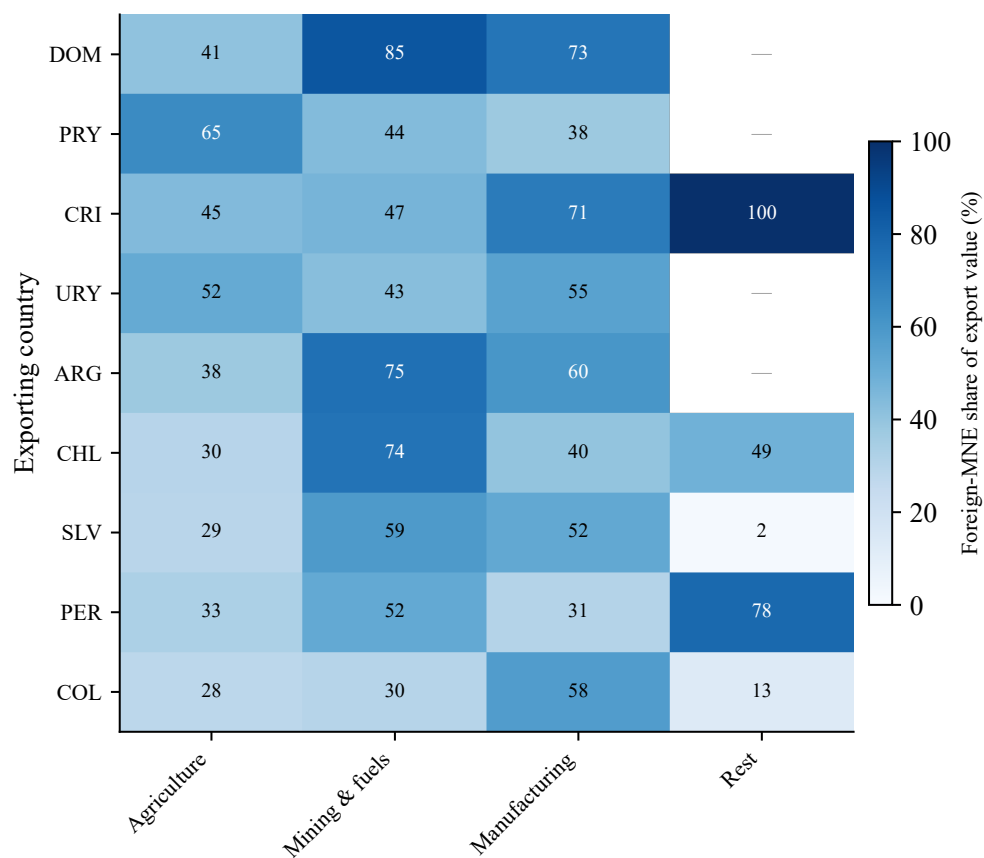


Figure 38: Foreign-Multinational Share of Export Value, by Exporting Country and Sector

Notes: Percent of each country's export value in each sector shipped by foreign MNEs.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 52: Exports by Exporting Country and Sector

Origin	Agriculture	Mining & fuels	Manufacturing	Rest	All
<i>Panel A: composition of the origin's exports, % (rows sum to 100)</i>					
ARG	57.4	9.9	32.7	0.0	100.0
CHL	21.9	29.7	46.9	1.4	100.0
COL	16.3	63.4	20.3	0.0	100.0
CRI	41.5	1.0	57.5	0.0	100.0
DOM	23.1	22.8	54.1	0.0	100.0
PER	19.4	61.0	19.6	0.0	100.0
PRY	85.2	1.1	13.7	0.0	100.0
SLV	22.1	3.0	74.9	0.0	100.0
URY	63.0	2.4	34.6	0.0	100.0
All origins	32.9	33.9	32.9	0.3	100.0
<i>Panel B: foreign-MNE share of export value, %</i>					
ARG	37.8	75.1	60.4	–	48.9
CHL	29.6	73.9	39.9	48.6	47.9
COL	28.0	30.1	57.8	12.7	35.3
CRI	44.5	46.9	70.9	100.0	59.7
DOM	40.7	85.3	73.2	–	68.4
PER	33.0	52.2	30.6	78.0	44.2
PRY	65.0	44.1	37.9	–	61.1
SLV	28.9	58.8	52.5	1.5	47.4
URY	51.8	42.9	55.3	–	52.8
All origins	38.1	49.2	51.6	47.9	46.3

Notes: Panel A: composition of each country's exports by sector (rows add to 100). Panel B: percent of each cell's export value shipped by foreign MNEs.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 53: Industry of Matched Exporting Affiliates

NAICS sector of the matched exporting affiliate	% of matched export value
goods-producing or unknown	82.4
information, finance, professional, holdings (51–56)	7.7
wholesale / retail trade (42–45)	7.4
other services (61–92)	1.8
transport, logistics (48–49)	0.6

Notes: Percent of matched export value, by NAICS sector of the exporting affiliate. Services are not observed in customs records; the table shows how much of goods exports is shipped by affiliates classified in service industries.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.

Table 54: Foreign-Multinational Exports by Sector and Destination Region

Sector / Destination region	Latin America	North America	Europe	Asia	Africa	Oceania	Rest of World	Total
Agriculture	31.1	13.8	20.2	28.1	6.1	0.4	0.4	30.7
Mining & fuels	14.3	14.6	24.7	45.5	0.5	0.2	0.2	36.0
Manufacturing	50.8	20.7	9.3	17.3	0.7	1.1	0.2	38.5
Rest	37.3	42.2	17.2	2.9	0.1	0.2	0.1	0.2

Notes: Percent of each sector's foreign-MNE exports, by destination region (rows add to 100). Last column: foreign-MNE exports. Values are annual averages in US\$ billion.

Source: Authors' calculations based on customs data matched to Orbis and Dun & Bradstreet.